The Heterogeneous Bank Lending Channel of Monetary Policy*

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Abstract

This paper develops a quantitative model to analyze the role of bank heterogeneity in the transmission of monetary policy through the bank lending channel. We calibrate the model to the euro area to capture two distinct forms of heterogeneity: ex-ante differences in loan pricing practices and ex-post variation in capital positions driven by idiosyncratic default risks. Consistent with empirical impulse responses, banks in fixed-rate economies experience severe net interest margin compression during monetary tightening as funding costs rise while income from legacy loans remains unchanged, leading to capital erosion and deeper lending contractions. The elasticity of new lending to monetary policy is approximately one-third larger in economies with fixed-rate loans. We uncover new trade-offs between monetary and macroprudential policy.

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1. Introduction

This paper develops a quantitative model to analyze the role that heterogeneity in banks' interest-rate risk exposure plays in the transmission of monetary policy. It is widely accepted that monetary policy transmits to the real economy, in part, through the *bank lending channel* (Bernanke and Gertler, 1995). The bank lending channel suggests that changes in central bank policies affect the economy by altering the availability of credit from banks to borrowers.

While the bank lending channel is well understood, banks may respond differently to monetary policy depending on their exposure to interest-rate risk, resulting in heterogeneous effects across banks. This exposure is heterogeneous in that banks in some regions, or that specialize in specific sectors, predominantly offer loans with a fixed interest rate, whereas others predominantly offer variable-rate loans.

The goal of this paper is to understand how banks' interest-rate exposure affects monetary policy transmission. Theoretically, we demonstrate an irrelevance result: heterogeneity in interest-rate risk exposure only matters insofar as there is also ex-post leverage heterogeneity across banks in the presence of capital constraints. Otherwise, monetary policy transmission is identical in fixed- and variable-rate banking systems. Quantitatively, we find that heterogeneity becomes significant only when a significant number of banks approach binding regulatory constraints—a situation that arises in our calibrated model due to portfolio adjustment costs and substantial ex-post heterogeneity from loan-default risks. Under these conditions, the elasticity of new lending to monetary policy is approximately one-third larger in economies with fixed-rate banking systems than in variable-rate economies. The divergence also extends to bank failure rates: tightening increases failures in fixed-rate economies while reducing them in variable-rate systems.

An early empirical literature has established that rebalancing following changes in monetary policy varies systematically across banks with different risk exposures and risk-bearing capacities. Early work by Kashyap and Stein (2000) documented how the strength of the bank lending channel depends on banks' characteristics. More recent work with improved identification has established that banks with low risk-bearing capacity—those with high leverage or low capital ratios—transmit changes in policy rates more strongly than well-capitalized banks (see, for instance, Jiménez et al., 2012;

Dell'Ariccia et al., 2017; Altavilla et al., 2020).[1] Likewise, banks with greater interest-rate risk exposure—those with a higher share of fixed-rate loans—exhibit stronger monetary policy transmission (Altunok et al., 2023).

While this empirical literature identifies important differences in bank responses to monetary policy shocks, structural models are essential for two complementary reasons. First, quantitative models are needed to understand aggregate responses: cross-sectional estimates explain differences in lending responses across banks, but these heterogeneous effects do not translate directly to aggregate responses. Second, empirical estimates do not allow for meaningful counterfactual analysis. For instance, models are required to quantify how monetary policy transmission would change if banks were homogeneous (identical replicas of each other), to measure the effect of heterogeneity on bank failure rates during tightening cycles, and to assess how the role of heterogeneity varies with the stringency of regulatory constraints and the magnitude of adjustment costs. The contribution of this paper is to build such a model.

We calibrate our model to the euro area—a region where interest-rate risk exposure heterogeneity is particularly pronounced: Banks in France, Germany, Belgium, and the Netherlands predominantly price loans at fixed rates, while those in Spain, Italy, Finland, and Portugal use variable rates. This institutional variation creates systematic differences in interest-rate risk exposures across countries. Moreover, due to market frictions, interest rate risk hedging remains modest and varies over time and across institutions, leaving most banks exposed to it.[2] Thus, a particularly pressing question for the European Central Bank (ECB) is whether this ex-ante heterogeneity translates into different responses across countries to changes in their monetary policy stance. Our model allows us to quantify precisely the importance of this ex-ante heterogeneity for aggregate outcomes, and under what monetary policy and regulatory conditions it becomes most consequential.

[1] Other references include Gambacorta and Mistrulli, 2004, Kishan and Opiela, 2000, and Holton and Rodriguez d'Acri, 2018.

[2] Financing long-term loans with short-term deposits exposes banks to interest rate risk (English et al. (2018); Gomez et al. (2021); Ampudia and den Heuvel (2022)). Empirical evidence indicates that larger euro area banks make greater use of derivatives to hedge interest rate risk than their U.S. counterparts (Begenau et al., 2015; Hoffmann et al., 2018). However, the overall scale of hedging remains modest: Hoffmann et al. (2018) and Guerrini and Rice (2025) document that European banks that actively engage in interest rate risk hedging typically offset only about 25% to 40% of on-balance-sheet exposures, leaving them exposed to interest rate risk even after hedging.

The model. Our quantitative framework models two banking systems—one with fixed-rate loans and another with variable-rate loans—each containing a distribution of banks that differ in their leverage due to idiosyncratic default shocks. Banks face convex loan origination costs, liquidity requirements, and capital requirements. The model is calibrated to euro area data for 2013-2023, targeting both aggregate moments and the dispersion in capital buffers across banks. This calibration discipline enables us to assess whether the observed heterogeneity in loan-pricing practices yields quantitatively significant differences in monetary transmission.

Within each banking system, there is a unit mass of atomistic banks. Each bank invests in central-bank reserves and loans. Banks are financed through a combination of short-term liabilities (insured deposits and wholesale debt) and retained earnings (internal equity). Banks face a risky maturity transformation problem since their deposit liabilities mature in one period, whereas loans are long-term and subject to default risk. The demand for loans and supply of deposits is determined in general equilibrium, but all the action is focused on banks.[3]

The loan portfolio comprises vintages of long-term loans. The issuance of new loans is subject to convex loan-origination costs. Once in the balance sheet, loan defaults are correlated according to the single risk factor model of Vasicek (2002). Upon a loan default, the bank loses the interest payments, recovers a fraction of the principal, and writes off the principal loss. Idiosyncratic credit risk induces ex-post heterogeneity. Reserves at the central bank are short-term (mature in one period) and pay an interest rate controlled by the central bank.

Banks face regulatory constraints designed to capture real-world prudential policy: In terms of liquidity regulation, banks must hold reserves above a fraction of their total short-term liabilities at any point in time. In terms of capital regulation, banks must comply with the minimum capital requirement. Banks that fail to comply with this requirement are forced into resolution.

Despite the richness in the sources of risks, portfolio variables, and regulatory environment, banks' policies are size-independent: the only relevant individual state variables are the leverage and the average interest rate of their loan portfolio. This parsimonious state space is sufficient to capture various relevant forms of heterogeneity

[3]The model also features entrepreneurs, households, and the government. Entrepreneurs have access to long-term investment projects requiring bank funding. Households consume, own the banks, and save in deposits. The government includes the central bank operations, the management of the deposit insurance scheme, and tax receipts.

observed in the empirical analysis. This tractability makes the model-data contrast particularly transparent.

Quantitative results. The calibrated model demonstrates that the interaction between ex-ante heterogeneity in interest-rate risk exposure and ex-post heterogeneity in leverage generates meaningful differences in monetary policy transmission. Fixed-rate banks experience capital erosion when rates rise, as their existing loan portfolios yield below-market returns. When combined with substantial dispersion in bank leverage—driven by idiosyncratic default shocks—this capital erosion pushes a meaningful fraction of highly leveraged institutions into binding regulatory constraints that amplify lending contractions. Variable-rate banks face the opposite dynamic: rising rates improve interest margins, bolstering capital positions and relaxing constraints.

This mechanism produces differential failure rates across systems. Monetary tightening increases failure probabilities in fixed-rate economies while reducing them in variable-rate systems, reflecting how interest rate changes affect bank capital through their existing loan portfolios. The role of ex-post heterogeneity is critical: when we counterfactually reduce idiosyncratic default risk, making all banks ex-ante identical replicas, the differences between fixed and variable rate systems shrink dramatically. Aggregate lending responses converge and failure rate differentials nearly disappear. This confirms that ex-ante heterogeneity in interest-rate risk exposure matters for monetary policy precisely because ex-post heterogeneity in leverage pushes banks to their borrowing limits, where the type of interest rate exposure becomes decisive.

These findings carry important implications for monetary policy in currency unions and for macroeconomic modeling. For the ECB, our results demonstrate that a single monetary policy generates meaningfully heterogeneous effects across member states—not just in the cross-section of banks, but in aggregate credit supply and financial stability outcomes at the country level. The magnitude of these differences depends on how close national banking systems operate to regulatory constraints.

Monetary policy and financial stability. Finally, our work highlights how monetary policy interacts with financial stability. First, a loosening of macroprudential policy during a monetary contraction, in the form of a reduction in capital requirements, improves financial stability as well as the difference in monetary transmission between fixed and variable-rate economies. Second, financial stability considerations may

explain why monetary policy tightening should proceed in a gradual and predictable way.

Related literature. Our work contributes to the literature on heterogeneous banks and monetary policy transmission by providing a quantitative assessment of when and why heterogeneity in banks' loan pricing and leverage matters for aggregate and cross-sectional lending responses to monetary shocks. Related studies focus on heterogeneity along other margins: Coimbra and Rey (2023) examine the risk-taking channel; Corbae and D'Erasmo (2021) study the effects of regulation on bank risk-taking; Rios-Rull et al. (2020) analyze capital requirements; Bianchi and Bigio (2022) investigate the credit channel with interbank frictions and deposit withdrawals; Jamilov and Monacelli (2025) introduce ex ante heterogeneity in banks' returns to study the distribution of bank size and income; and Bellifemine et al. (2022) extend this framework with nominal frictions. Varraso (2025) studies monetary transmission when intermediaries invest in both short- and long-term assets. Relative to these contributions, our paper introduces a relevant dimension of heterogeneity—loan pricing—and shows that its interaction with leverage heterogeneity is central to shaping monetary policy transmission.

We depart from the classical representative-bank macroeconomic literature along several key modeling assumptions. First, rather than assuming that banks directly hold productive assets—as in Gertler and Kiyotaki (2010), He and Krishnamurthy (2013), Brunnermeier and Sannikov (2014), or Nuno and Thomas (2017) among others—we explicitly model loan contracts between borrowers and banks. Second, our framework accounts for default risk in banks' loan portfolios, with defaults correlated within each bank according to Vasicek (2002) single risk factor model. This captures the asymmetric risk profile of defaultable loans, characterized by limited upside but unlimited downside risk (Nagel and Purnanandam, 2020; Mendicino et al., 2024). Third, we treat capital requirements as occasionally binding constraints, in contrast to previous models that assume always-binding capital constraints (see, e.g., Boyarchenko and Adrian, 2015; Clerc et al., 2015; Mendicino et al., 2018, 2020).

A related strand examines monetary transmission through the mortgage market. Berger et al. (2021) and Eichenbaum et al. (2022) emphasize the path-dependency of policy rates in shaping household consumption. Greenwald (2018) highlights the role of loan-to-valuation and payment-to-income constraints, while Beraja et al. (2018) focus

on the importance of home equity.[4] In the euro area, cross-country heterogeneity in monetary transmission to economic aggregates is studied by Corsetti et al. (2021), while Calza et al. (2013) emphasize the role of housing finance. More recently, Pica (2022) and Sciacovelli (2025) focus on the role of adjustable-rate mortgages. These works model heterogeneous households facing uninsurable income risk, transaction costs, and borrowing constraints, while abstracting from lender heterogeneity and taking a simplified approach of credit supply.

Finally, our paper relates to work on how interest rate risk exposure affects monetary policy transmission through banks. Begenau et al. (2024) investigate the financial stability implications of banks' interest rate risk and uninsured deposit funding. Wang (2025) studies how prolonged declines in interest rates affects bank intermediation spreads and credit supply. Guren et al. (2021) and Elenev and Liu (2025) examine how mortgage contract design—fixed versus adjustable rates—shapes macroeconomic volatility, household default risk, and housing demand. In contrast, our work shows how interest rate risk arising from banks' loan pricing practices shapes the allocation of risk between borrowers and banks in response to monetary shocks.

2. The model

We consider an infinite-horizon, discrete-time economy, where time is indexed by $t \in \{0, 1, 2, \ldots\}$ and there is a single good. The economy is populated by four types of agents: a representative household, a mass of entrepreneurs, a continuum of competitive banks, and a consolidated government. The government consists of a bank regulatory authority, a deposit insurance agency, and fiscal and monetary authorities.

The core of the model is a banking sector that intermediates funds from households to entrepreneurs. Entrepreneurs undertake risky, long-term productive investment projects that require external financing. Banks engage in maturity transformation, funding these long-term loans with short-term retail deposits, wholesale debt, and equity. This activity exposes banks to credit risk and interest rate risk.

The regulatory framework, comprising capital and liquidity requirements and a deposit insurance scheme, constrains banks' decisions and shapes their response to aggregate shocks. Aggregate economic activity depends on the interaction between

[4] Kaplan et al., 2018; Auclert, 2019; Garriga and Hedlund, 2020 examine monetary policy transmission in heterogeneous-agent economies.

banks' lending capacity and entrepreneurs' investment demand.

We analyze two distinct institutional arrangements regarding loan-rate fixation: one where loan contracts stipulate a fixed interest rate for the life of the loan, and another where the interest rate is variable, resetting each period. We refer to these two setups as the *fixed-rate (FR) economy* and the *variable-rate (VR) economy*, respectively. This distinction allows us to isolate how the exposure to interest-rate risk affects the banking sector and, in turn, macroeconomic outcomes. The following subsections detail the objectives, constraints, and technology available to each agent.

2.1 Banks

The banking sector consists of a continuum of ex-ante identical, perfectly competitive banks, indexed by $j \in [0,1]$. Banks operate under limited liability and are managed by risk-neutral bankers with a subjective discount factor $\beta \in (0,1)$ who maximize the discounted value of dividends for their owners, the households.

Banks finance a portfolio of risky long-term loans and safe short-term assets using a combination of short-term insured deposits, wholesale debt, and equity accumulated via retained earnings.

Assets. Bank assets comprise risky long-term loans and safe short-term assets, which we refer to as reserves.[5] At the beginning of period t, bank j holds a portfolio of legacy loans, L_{jt} originated in previous periods. It then chooses its origination of new loans, N_{jt}, and its holdings of central bank reserves, M_{jt}.

Reserves, M_{jt}, are a risk-free, one-period asset that pays a net interest rate r_t^M, which is the policy rate set by the monetary authority.

The loan portfolio consists of a continuum of long-term loans, each with a principal normalized to one. Following Leland and Toft (1996), each loan matures with an i.i.d. probability $\delta \in (0,1)$, implying an average loan maturity of $1/\delta$. Loans are subject to credit risk, which we model using the single-risk-factor framework of Vasicek (2002), described in Appendix A.1. The fraction of a bank's loan portfolio that defaults, ω_{jt+1}, is a random variable drawn from a time-invariant distribution $F(\omega)$ with mean $\mathbb{E}[\omega] = p \in [0,1]$. Upon default, the bank recovers a fraction $1-\lambda$ of the principal, where $\lambda \in [0,1]$ represents the loss given default.

[5] These assets can be thought of as central bank reserves or as safe short-term government bonds.

The law of motion for the bank's legacy loan portfolio is:

$$L_{jt+1} = (1-\omega_{jt+1})(1-\delta)(L_{jt}+N_{jt}). \tag{1}$$

This formulation implies that the portfolio at $t+1$ consists of the previous period's total loans, $L_{jt}+N_{jt}$, net of maturing and defaulted loans.

Origination of new loans incurs a convex cost $f(N_{jt}/L_{jt})L_{jt}$, where $f(\cdot)$ is an increasing and convex function. This convexity captures the increasing marginal difficulty of finding creditworthy borrowers or screening profitable investment opportunities as the bank expands its lending relative to its existing customer base.

The contractual interest rate of a bank's loans depends on the institutional environment. The interest rate on new loans originated at time t is denoted r_t^N.[6] In the FR economy, the net interest rate r_t^N is fixed at origination and remains constant for the life of the loan. In the VR economy, what is fixed at origination is the spread s_t^N, which is added to the policy rate r_t^M set by the monetary authority, such that the rate is $r_t^N = r_t^M + s_t^N$. Hence, in this case, the contractual spread remains constant for the life of the loan, but the interest rate fluctuates over time with the policy rate.

For FR banks, the average interest rate on a bank's legacy loan portfolio, r_{jt}^L, evolves according to:

$$r_{jt}^L = \frac{r_{jt-1}^L L_{jt-1} + r_{t-1}^N N_{jt-1}}{L_{jt-1}+N_{jt-1}}. \tag{2}$$

For VR banks, the return is $r_{jt}^L = r_t^M + s_{jt}^L$, where the average contractual spread s_{jt}^L follows the law of motion:

$$s_{jt}^L = \frac{s_{jt-1}^L L_{jt-1} + s_{t-1}^N N_{jt-1}}{L_{jt-1}+N_{jt-1}}. \tag{3}$$

Liabilities. The bank's assets are funded with a combination of wholesale debt B_{jt}, (retail) deposits D_{jt}, and equity E_{jt}. Wholesale debt is a one-period liability that pays a net interest rate r_t^B. Retail deposits are one-period liabilities paying r_t^D. They provide liquidity services to depositors (which implies that, in equilibrium, $r_t^D \leq r_t^B$). A bank's

[6] Note that, given our perfect-competition assumption, banks are price-takers in the loan market, making this rate the same for all banks in a given period and thus not indexed by j.

ability to issue deposits is constrained by the size of its legacy loan portfolio:

$$D_{jt} \leq \alpha L_{jt}, \tag{4}$$

with $\alpha \leq 1$. We interpret this constraint as capturing the specific nature of relationship banking: deposits are often a by-product of lending relationships, as borrowers are required to open accounts or maintain balances as part of their loan covenants. Alternatively, this can be viewed as a reduced-form representation of the synergies between loan origination and deposit taking, such as the shared physical branch network required for both activities.

We assume that retail deposits are fully insured by the government and that, while wholesale debt is not, its returns are also risk-free in equilibrium.[7]

Banks accumulate equity exclusively through retained earnings (i.e., we assume there is no external equity issuance). The law of motion for equity is:

$$E_{jt+1} = E_{jt} + (1-\tau)\Pi_{jt+1}, \tag{5}$$

where $\tau \in (0,1)$ is the corporate tax rate. Π_{jt+1} denotes pre-tax profits realized between period t and $t+1$:

$$\begin{aligned}\Pi_{jt+1} = (1-\omega_{jt+1})\left(r^L_{jt}L_{jt} + r^N_t N_{jt}\right) + r^M_t M_{jt} - r^D_t D_{jt} - r^B_t B_{jt} \\ - \lambda\omega_{jt+1}(L_{jt}+N_{jt}) - f\left(\frac{N_{jt}}{L_{jt}}\right)L_{jt} - \bar{\pi}L_{jt}.\end{aligned} \tag{6}$$

where the first line is the net interest income—the difference between the interest earned on assets and the interest paid on liabilities—and the second line includes realized credit losses, loan origination costs, and operational costs, which are a constant fraction $\bar{\pi} > 0$ of the legacy loan portfolio.

The balance sheet of the bank is:

$$L_{jt} + N_{jt} + M_{jt} = D_{jt} + B_{jt} + E_{jt}. \tag{7}$$

[7] To obtain this result, we need to assume that wholesale debt is either senior to deposits, or that it is collateralized with the bank's assets. This imposes some parametric restrictions in terms of the relative size of each of these sources of funding and/or the recovery value of a bank's assets in case of default, such that wholesale debt returns are effectively risk free (see Appendix A.2 for a derivation of those restrictions).

Regulation. The banking system is subject to both liquidity and capital regulation, akin to the Basel III framework. Liquidity regulation imposes a minimum amount of reserve holdings proportional to the bank's short-term liabilities:

$$M_{jt} \geq \theta(D_{jt} + B_{jt}). \tag{8}$$

Capital regulation imposes that a bank's equity must cover at least a fraction $\gamma \in (0,1)$ of its total loan portfolio:

$$E_{jt} \geq \gamma(L_{jt} + N_{jt}). \tag{9}$$

Bank failure, entry and exit. A bank fails and is resolved by the regulator if, after the realization of portfolio defaults ω_{jt+1}, its equity falls below regulatory minimum: $E_{jt+1} < \gamma L_{jt+1}$. Upon failure, its equity is wiped out, and a deposit insurance agency seizes its assets, liquidates a fraction, and sells the remainder to new entrants. The agency allocates is proceeds to the bank's liability holders, in order of seniority, and repays all retail depositors in full.

Additionally, banks face an exogenous exit shock with probability $\chi \in (0,1)$ each period. Exiting banks repay liabilities and distribute remaining equity as dividends. To maintain a constant mass of banks, each exiting bank is replaced by a new entrant. New entrants start with an exogenous amount of equity $\bar{E}_t$ and a random amount of legacy loans that ensures that the leverage distribution of new banks is the same as that of surviving banks. These exit and entry dynamics ensure a stationary distribution of bank sizes. The fraction of loans in the legacy loan portfolio of exiting banks at $t+1$ that is not distributed among new banks, which we denote $\tilde{\chi}$, is liquidated.[8]

Recursive formulation. The state of an individual bank j at time t is summarized by its legacy loans L_{jt}, equity E_{jt}, and the average interest rate on its legacy portfolio r^L_{jt}, for FR banks, or the average spread s^L_{jt}, for VR banks. The bank's value function V_t

[8]We fix the amount of equity of entering banks $\bar{E}$ in the steady state to normalize the aggregate size of the banking sector. Given this parameter value, we can calculate the implied steady-state value of $\tilde{\chi}$. In response to shocks $\bar{E}_t$ adjusts such that the implied $\tilde{\chi}$ remains constant and equal to its steady state value.

satisfies:

$$V_t\left(L_{jt}, E_{jt}, x^L_{jt}\right) = \mathbf{1}_{\{E_{jt} \geq \gamma L_{jt}\}}\Bigg[\max_{\{N_{jt}, M_{jt}, D_{jt}, B_{jt}\}} \beta \mathbb{E}_t\Big[(1-\chi) V_{t+1}\left(L_{jt+1}, E_{jt+1}, x^L_{jt+1}\right) + \chi E_{jt+1}\Big]\Bigg], \tag{10}$$

with $x_{jt} = r^L_{jt}$ for FR banks and $x_{jt} = s^L_{jt}\}$ for VR banks. The optimization problem of the bank is subject to the laws of motion for loans (1) and for average legacy rate (2) for FR banks, or the average legacy spread (3) for FR banks, the constraint on retail deposits (4), the law of motion for equity (5), the balance-sheet constraint (7), and the regulatory constraints (8) and (9). The indicator function captures the failure condition. Lemma 1 below shows that the problem can be written more parsimoniously in terms of two state variables: the bank's leverage (L_j/E_j) and either the average legacy rate (r^L_j) for FR banks, or the average legacy spread (s^L_j) for VR banks.

2.2 Entrepreneurs: Loan demand microfoundation

A mass of risk-neutral entrepreneurs, indexed by $i \in [0,1]$, have access to an investment technology requiring the upfront use of one unit of the final good. Entrepreneurs are endowed with no internal funds and must obtain a bank loan to finance their projects.

An active project yields A units of the final good per period. At the end period t, the project terminates if: (i) it reaches successful completion, which occurs with probability δ; (ii) it fails, which occurs with probability p; or (iii) its loan is liquidated as a result of the exit of its financing bank, which occurs with probability $\tilde{\chi}$. If the project is completed or the loan is liquidated, the principal is repaid in full. If the project fails, the bank recovers only $1-\lambda$ of the principal.

Initiating a project requires a utility cost of $a(N_t)$, where $a(\cdot)$ is an increasing and convex function of N_t, the aggregate volume of new projects. This cost generates an upward-sloping supply curve for new projects, which captures aggregate decreasing returns to scale for the entrepreneurial sector. Free entry for entrepreneurs implies that, in equilibrium, the expected lifetime value of a new project must equal this startup cost. This condition implies a uniform interest rate r^N_t for all new loans originated at time t.

The value of a project depends on the interest rate regime. For a VR loan, the value

at time t of a project with spread s_i^N is:

$$V_{it}^E(s_i^N) = \sum_{k=0}^{\infty} \beta^{k+1}(1-p)^{k+1}(1-\delta)^k(1-\tilde{\chi})^k \left[A - \left(r_{t+k}^M + s_i^N\right)\right], \tag{11}$$

The free-entry condition $V_{it}^E(s_t^N) = a(N_t)$ implies aggregate demand for VR loans:

$$N_t = a^{-1}\left(\sum_{m=0}^{\infty} \Omega_m \left[A - \left(r_{t+m}^M + s_t^N\right)\right]\right), \tag{12}$$

where $\Omega_m \equiv \beta^{m+1}(1-p)^{m+1}(1-\delta)^m(1-\tilde{\chi})^m$ is the entrepreneurs' effective discount factor. Loan demand in the VR economy is forward-looking, as entrepreneurs form expectations about future interest rates.

For a FR loan, the interest rate r_t^N is fixed for the loan's duration. Aggregate loan demand becomes:

$$N_t = a^{-1}\left(\sum_{m=0}^{\infty} \Omega_m \left(A - r_t^N\right)\right). \tag{13}$$

In this case, loan demand is not forward-looking with respect to future interest rates, as the bank bears all interest-rate risk. This distinction drives the differential transmission mechanism in our analysis.

2.3 Households: supply of bank funds

The household block determines the supply of funds available to banks. It plays a dual role. First, unlike partial equilibrium or ad-hoc formulations common in the literature, embedding household decisions in general equilibrium disciplines the analysis: it ensures that a consistent representation of funding supply exists and forces us to be explicit about how monetary policy is implemented. Second, the specific modeling choices ensure tractability: they deliver a static demand system—depending only on current rates—while government bond supply shocks, occurring contemporaneously with monetary policy shocks, allow both the deposit rate and the rate on reserves to follow their observed empirical trajectories. The loan rate, in contrast, responds endogenously. This flexibility to match the empirical path of funding costs while preserving endogenous loan pricing is key to accurately capturing banks' cost of funds, allowing us to squarely focus on the bank-lending channel.

With that purpose in mind, households have quasi-linear preferences over two

consumption goods—one entering with curvature, the other linearly—and derive additional utility from holding a bundle of monetary assets. The bundle combines highly liquid assets, D_t^H, with less liquid bonds, A_t^H, and the differing liquidity services of these assets generate an equilibrium spread between their returns. Highly liquid assets comprise bank deposits and short-term government paper; bonds comprise wholesale debt issued by banks, B_t^H, and longer-term government bonds, M_t^H. Within each category, the component assets are perfect substitutes from the household's perspective. Appendix A.3 details the full problem.

The core result is a canonical asset-demand system. In particular, the demands for highly liquid assets and bonds are:

$$D_t^H = h^D(r_t^D, r_t^M), \tag{14}$$

and

$$A_t^H = h^A(r_t^D, r_t^M), \tag{15}$$

where $h^D(\cdot)$ and $h^A(\cdot)$ are the respective demand functions. Perfect substitutability between wholesale debt and government bonds implies that only two rates enter this system. Because deposits provide greater liquidity services, in equilibrium $r_t^D \leq r_t^M$.

The static demand system, which simplifies computation, follows directly from our assumptions. Quasi-linear preferences—standard in new-monetarist models (Lagos and Wright, 2005; Lagos et al., 2017) and also used in dynamic banking models (Bianchi and Bigio, 2022)—deliver demand functions that, unlike loan demand, depend only on current rates. The complementarity between liquid assets and bonds couples the two demands into a system, as in other quantitative models with competing monetary services (Drechsler et al., 2017; Di Tella and Kurlat, 2021), preventing a perfect pass-through from policy rates to deposit rates.

2.4 Consolidated government

The consolidated government includes a central bank and a fiscal authority. As is standard, the central bank supplies reserves to implement the policy rate r_t^M. The fiscal authority raises taxes from banks and households and manages the deposit insurance scheme. In addition, the government issues short-term bonds that, from the household's perspective, are perfect substitutes for deposits. Adjusting the supply of

these bonds shifts the household demand system derived above, allowing the model to match the empirical response of the deposit rate r_t^D to monetary policy shocks. In the quantitative analysis, we treat both the policy rate and the deposit rate paths as exogenous, with reserve and bond supplies adjusting in the background to implement them.

These operations are consolidated in the following government budget constraint:

$$T_t + \tau \Pi_t + M_t^S + D_t^S = \left(1 + r_{t-1}^M\right) M_{t-1}^S + \left(1 + r_{t-1}^D\right) D_{t-1}^S + \Upsilon_t, \tag{16}$$

where T_t denotes lump-sum taxes paid by households, Π_t aggregate profits from banks, M_t^S the supply of reserves, D_t^S the supply of short-term government bonds, and Υ_t the net operating deficit of the deposit insurance scheme.

2.5 Equilibrium and characterization

Definition 1. *An equilibrium is a sequence of prices and allocations such that:*

1. *Banks maximize value subject to regulatory and balance sheet constraints, taking prices* $\{r_t^N, s_t^N, r_t^M, r_t^B, r_t^D\}$ *as given.*
2. *Entrepreneurs' aggregate credit demand satisfies the free entry condition.*
3. *Households optimize consumption and asset holdings.*
4. *Markets for new loans, deposits, wholesale debt, public liquidity, and consumption goods clear.*
5. *The government budget constraint holds.*

Appendix A.4 provides more details on all equilibrium objects.

Size independence. A key feature of our model is that banks' optimal policies exhibit size independence, which substantially simplifies the analysis while preserving the essential heterogeneity. It implies that banks' decision rules can be expressed independently of their current equity level. Specifically, we can factor out current equity from all policy functions, leaving leverage—the ratio of legacy loans to equity—and the average rate/spread of legacy loans as the sole relevant state variables for individual bank decisions.

Lemma 1 (Size independence)**.** *The bank's value function* $V_t\left(L_{jt}, E_{jt}, x^L_{jt}\right)$*, which solves* (10) *is linear in bank equity*

$$V_t\left(L_{jt}, E_{jt}, x^L_{jt}\right) = v_t\left(l_{jt}, x^L_{jt}\right) E_{jt}.$$

where

$$v_t\left(l_{jt}, x^L_{jt}\right) = \max \beta \mathbb{E}_t \left[(1-\chi)\, v_{t+1}(l', x^{L'}) \frac{E'}{E} + \chi \frac{E'}{E}\right]$$

Proof. See Appendix A.5.

This homogeneity arises because returns and costs scale linearly with the balance sheet, while regulatory constraints are proportional to assets. Consequently, bank growth depends on leveraged returns and idiosyncratic shocks, not absolute scale. We characterize the economy by tracking the distribution of leverage and, during transitions, the average loan rate/spread. This simplification reduces the dimensionality of the problem considerably, while preserving the crucial interactions between leverage, regulatory constraints, and loan pricing that drive the heterogeneous transmission of monetary policy.

Irrelevance of interest-rate risk exposure. A central theme of this paper is that differences in the transmission mechanism of monetary policy in FR and VR economies follow predominantly from the interaction between monetary policy and capital regulation. We provide a theoretical benchmark establishing conditions under which the transmission of monetary policy is identical in FR and VR economies.

Proposition 1 (Irrelevance of interest-rate risk exposure)**.** *Given a sequence of policy rates* $\{r^M_t\}$ *and deposit rates* $\{r^D_t\}$*, and starting from identical initial states, the equilibrium paths of aggregate lending in the FR and VR economies coincide if:*

1. *Capital requirements are not binding (i.e., constraint* (9) *holds with strict inequality) for all banks at all times.*

2. *Banks and entrepreneurs discount future cash flows at identical rates.*

Proof. See Appendix A.6.

The logic underlying Proposition 1 mirrors that of the classical Modigliani-Miller theorem (Modigliani and Miller, 1958), which states that the structure of liabilities is irrelevant for investment decisions when the discount rates of the firm and its sources of finance are equalized. Our result offers an analogous irrelevance: the structure of loan contracts—specifically, how loan contracts allocate interest rate variation between borrower and lender—is immaterial when entrepreneurs bear the marginal return on investment and banks bear the marginal cost of funding, provided both parties discount future cash flows at the same rate.

The key insight is that, regardless of whether interest rates are fixed or variable, the *total* present value of interest payments over the life of a loan depends only on the discounted sum of the relevant rates—either the fixed rate r_t^N or the sequence of variable rates $\{r_{t+m}^M + s_t^N\}_{m\geq 0}$. When banks and entrepreneurs share a common discount factor (Condition 2), the scale of projects is determined solely by equating the marginal return to new projects minus the marginal cost of those projects, which combines adjustment costs and banks' costs of funds. This condition ensures that the discounted net present value (NPV) of interest payments is identical under both arrangements, implying that the choice between FR and VR becomes a matter of *how* the surplus of funding new projects is split over time.

Condition 1—that capital requirements remain non-binding—plays a central role in guaranteeing that banks in FR and VR economies apply the same discount factor as entrepreneurs. When regulatory constraints do not bind, banks' lending decisions are governed solely by the NPV trade-off between interest income and funding costs. The fixed-versus-variable distinction affects only the *timing* of when interest rate movements impact bank profits, not the present value of those profits. A VR bank experiences immediate pass-through of policy rate changes to its legacy portfolio, while a FR bank faces gradual adjustment as new loans replace maturing ones. However, when both types of banks discount future cash flows identically and face no binding constraints, these timing differences wash out in present value terms.

Condition 2—the alignment of discount factors—holds exactly when banks and entrepreneurs faced identical tax treatment, exit rates, and access to risk-free asset investments. In our full model, these features introduce wedges between the discount factors of banks and entrepreneurs. The irrelevance result obtains analytically once we abstract away from these features. We retain the differential tax treatment because it is quantitatively important for matching the empirical moments documented in Section 3

and we prevent entrepreneurs from investing in the risk-free asset to isolate the effects of interest rate changes on loan supply, while keeping demand fixed. Nevertheless, we emphasize later that departures from Condition 2 are not the primary driver of the differences we document. As we demonstrate in Section 4, the quantitative differences between FR and VR economies remain modest when idiosyncratic risk is reduced, even though Condition 2 is not satisfied exactly.

The requirement that capital constraints do not bind admits a natural interpretation in terms of the model's risk calibration. Recall that ex-post heterogeneity in bank leverage arises from idiosyncratic loan default shocks—as $\rho \to 0$, idiosyncratic risk vanishes. In this limit, provided that parameters are chosen such that the representative bank maintains adequate capital buffers, Condition 1 is satisfied. The proposition thus clarifies that differences in monetary policy transmission between FR and VR economies emerge fundamentally from the *interaction* between interest-rate risk exposure and financial constraints, rather than from interest rate exposure per se. We return to this condition after calibrating the model.

2.6 Solution method

To solve the model, we use a value function iteration algorithm for the bank problem defined in Appendix A.5 and keep track of the bank distribution over log-equity $\log(E_{jt})$, leverage $l_{jt} = \frac{L_{jt}}{E_{jt}}$, and the loan rate/spread x_{jt} using the method of Young (2010). The steady state can then be found in an iterative procedure where, for a given guess of the loan rate r^N, we first solve for the policy functions using value function iteration and then compute the bank distribution using those policy functions. During this procedure, the guess for the loan rate is adjusted until an equilibrium in the loan market is found. The transitional dynamics after an MIT shock are computed similarly to Boppart et al. (2018). For a guess of a transition path for the loan rate $\{r_t^N\}_{t=0}^T$, we make a backward pass along the transition to compute the policy functions, followed by a forward pass to compute the distribution along the transition. Similar to our steady state algorithm, we adjust the transition path for the loan rate $\{r_t^N\}_{t=0}^T$ until the loan market clears in each period. Appendix C provides more details on the solution algorithm.

3. Calibration and model fit

We calibrate the model to the euro area economy at a quarterly frequency. Subsection 3.1 specifies functional forms adopted in the calibration, subsection 3.2 presents parameter values, and subsections 3.3 and 3.4 evaluate the model fit along the cross-sectional and time-series dimensions, respectively.

3.1 Functional forms

Loan-origination cost. We assume a quadratic cost function for originating new loans:

$$f(N_{jt}/L_{jt}) = \eta\left(\frac{N_{jt}}{L_{jt}}\right)^2, \quad \eta > 0. \tag{17}$$

Credit risk. We model the cumulative distribution function (cdf) of the loan default rates ω_{jt+1} as in the Vasicek (2002) single risk-factor model (see Appendix A.1):

$$F_j(\omega) = \Phi\left(\frac{\sqrt{1-\rho}\Phi^{-1}(\omega) - \Phi^{-1}(p)}{\sqrt{\rho}}\right), \tag{18}$$

where $\Phi(\cdot)$ is the cdf of a standard normal, $\Phi^{-1}(\cdot)$ denotes its inverse, and $\rho \in [0,1]$ is the so-called correlation parameter, which measures the strength of the dependence of individual loan defaults on the common risk factor.[9]

Entrepreneurs' entry cost. The cost for entrepreneurs to initiate a project is:

$$a(N_t) = \zeta_1 N_t^{\zeta_2}, \tag{19}$$

with $\zeta_1, \zeta_2 > 0$. These parameters govern the scale of the loan demand and its interest-rate semi-elasticity, respectively.

3.2 Calibration

The calibration strategy consists of a two-step procedure. In the first step, a number of parameters are set by either following the calibration of other papers or fixing them

[9] See, e.g., Repullo and Suarez (2004) for details. This distribution was used as a statistical foundation for the capital requirement formulas of the IRB approach when first introduced by Basel II (Gordy, 2003).

directly to their observed regulatory counterparts. In the second step, values for the remaining parameters are set so as to jointly match a number of empirical moments using euro area data. Table 1 reports all parameter values corresponding to both groups.

Pre-set parameters. The first block of Table 1 corresponds to the pre-set parameters. We follow Mendicino et al. (2020) and set the average loan default rate p to 2.65% (annualized) and the loan-loss given default λ to 0.3. The average maturity of loans is set to 0.05, consistent with an average loan duration of 5 years reported by Cortina et al. (2018), which corresponds to the maturity observed for syndicated loans in developed economies from 1991 to 2014. The corporate tax rate τ is set at the average effective tax rate of 20% for European banks.[10]

Policy parameters are set based on Basel III regulatory levels: The capital requirement γ encompasses the minimum Common Equity Tier 1 (CET1) requirement, 4.5%, plus a capital conservation buffer, 2.5%, that must also be maintained with CET 1 capital. The parameter α in equation (4), which determines the share of deposits in banks' balance sheet, is set to match the observed ratio of deposits-to-total-assets of 0.78, while the liquidity requirement θ matches the reserve-to-total-liabilities ratio of 0.118, both ratios are consistent with the consolidated balance sheet of monetary financial institutions in the euro area.[11] The steady-state value of the policy rate r^M (i.e., the rate of remuneration on central bank reserves) is set to the historical average of the nominal deposit facility rate in the euro area (1%). Similarly, the steady-state deposit rate r^D is set to 0.5% according to the historical average of the nominal overnight deposit rate in the euro area.

Jointly calibrated parameters. The second block of Table 1 corresponds to parameters calibrated jointly to match a number of targeted moments. The targets for banks' return on equity (ROE) of 6.4 percent and the bank failure probability of 0.66 percent are taken directly from Mendicino et al. (2020). These targets are key to identifying the bankers' discount factor β and the volatility of a bank's portfolio default rate ρ, respectively. Similarly, the scale parameters η of the loan-origination cost in (17), ζ_2 and ζ_1 of the

[10] See the Damoradan database http://www.stern.nyu.edu/~adamodar/pc/datasets/taxrateEurope.xls

[11] See Appendix B, Figure 11, for a detailed composition of MFIs and a description of the time series used. We cannot distinguish MMF from the deposit-taking institutions for the entire time series, but this does not distort the data much as the size of MMFs is very small compared to deposit-taking institutions. In 2024Q2, for instance, MMF aggregate size was 1.8 tn euros, less than 5% the size of deposit-taking institutions, which amounts to 38 tn.

Table 1: Parameter values

Pre-set parameters

	Parameter	Value	Target/Source
p	Loan default rate, mean (%)	0.6625	Mendicino et al. (2020)
λ	Loan loss-given-default	0.30	Mendicino et al. (2020)
δ	Loan maturity	0.05	Cortina et al. (2018)
τ	Corporate tax rate	0.20	Damodaran database.
γ	Min. capital requirement (%)	7.0	Basel III CET1 + Buffer requirement.
α	Deposits-to-loans ratio	0.78	Consolidated EA banks balance sheet.
θ	Liquidity requirement (%)	11.8	Liquid asset to total deposit ratio.
r^M	Steady-state policy rate (%)	1.0	Avg. EA deposit facility rate, 1999-2019.
r^D	Steady-state deposit rate (%)	0.5	Avg. EA overnight deposit rate, 2003-2023.

Jointly calibrated parameters

	Parameter	Value	Target	Data	Model
β	Subjective discount factor	0.933	Banks' return on equity (%)	6.4	6.4
ρ	Loan default correlation	0.51	Bank failure probability (%)	0.66	0.66
η	Loan origination cost	0.22	Voluntary capital buffer (%)	5.1	4.8
ζ_1	Ent. entry cost (level)	5.78	Avg. lending rates (%)	3.0	3.0
ζ_2	Ent. entry cost (power)	0.50	Response of new lending (%)	-0.38	-0.37
$\bar{\pi}$	Fixed operating cost	0.012	Non-interest expenses to assets (%)	0.34	0.22
χ	Bank's exit rate (pp)	2.00	Slope of log-log asset distribution	-1.56	-1.56

Note: Interest rates and probabilities are reported in annualized terms.

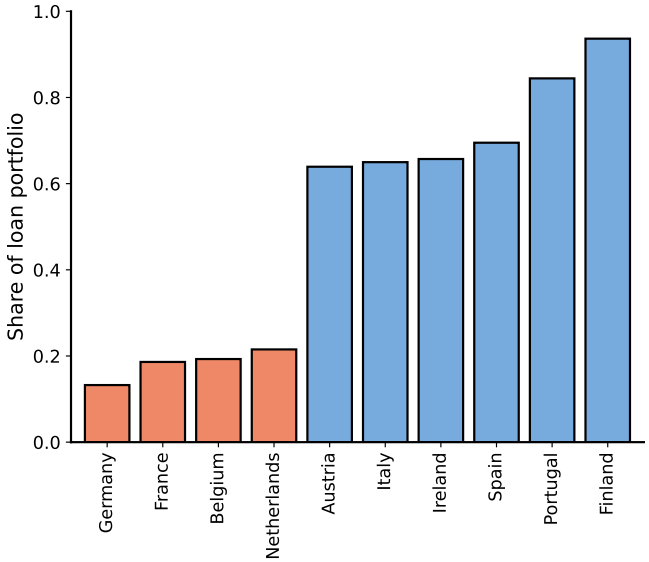
loan demand in (19) are jointly estimated to target the historical average loan rates of 3%, the peak response of the logarithm of new lending, -0.38, to a monetary policy shock of 100 basis points (see Figure 3 below), and the average voluntary capital buffer of 5.1 percent, a target consistent with the mean CET1 buffer for supervised banks reported by the European Banking Authority for 2021.[12]

The bank's fixed operating cost parameter $\bar{\pi}$ is informative about the average net

[12]We calibrate the model based on the CET1 estimates for Supervised banks, as it represents the most accurate available estimate for capital buffers. See Appendix B.3 for a comparison of different estimates of CET 1 ratios and buffers.

non-interest expenses-to-assets ratio, and the bank's exit rate χ is disciplined by targeting the slope of the log-log regression coefficient (tail coefficient) of the asset-size distribution. As discussed in Section 3.3, this moment allows the model to replicate a power-law distribution of bank asset sizes, a salient feature of the data.

Figure 1: Share of variable-rate loans.



Note: Average share of the aggregate total outstanding loans issued at variable rates from 2014 to 2020. Includes loans to non-financial corporations and loans to households– mortgage loans, consumer loans, and other loans. Orange bars corresponds to our classification of fixed-rate countries and Blue bars to variable-rate ones. *Source*: ECB MFI Statistics.

Ex-ante heterogeneity: FR and VR economies. As anticipated above, we study two versions of the model, one with FR and the other with VR. This feature is motivated by cross-country institutional patterns in the observed countries of the euro area. Figure 1 presents the share of VR loan contracts in each country. VR loans are defined as those with original and remaining maturity over 1 year and interest rate reset within the next 12 months.[13] We observe that in a first group of countries, consisting of Germany, France, Belgium, and the Netherlands, around 80% of outstanding loans correspond

[13]At the country-consolidated level, the time series for loans are not exclusively categorized by the type of interest rate fixation but rather by their maturity. We also approximate the share of variable-rate loans by categorizing loans with a maturity of up to a year, which produces similar results. The categorization in Figure 1 aligns with the results reported by Core et al. (2025) using granular-level data for non-financial corporate loans in the euro area.

to FR contracts. In contrast, in a second group, Finland, Portugal, Spain, Ireland, Italy, and Austria, above 60% of outstanding loans correspond to VR contracts.[14] We label the first and second group as FR and VR banking systems, respectively. We employ these definitions in Section 4, where we study the aggregate response of banks' balance sheet variables to monetary shocks by country groups.

3.3 Cross-sectional validation

We validate the model by comparing untargeted cross-sectional moments with the data.

Bank balance sheet composition. We begin by comparing the consolidated balance sheet of monetary financial institutions (MFIs) operating in the euro area to our model's counterpart.[15] Table 2 shows that the model's steady-state consolidated balance sheet closely aligns with the composition of assets and liabilities in the data.

Table 2: Consolidated bank balance sheet: Model vs. data (2013–2023)

Assets			**Liabilities**		
	Model	Data		Model	Data
Loans	88%	88%	Deposits	81%	78%
ST securities and reserves	12%	12%	Wholesale funding	9%	14%
			Equity capital	10%	8%

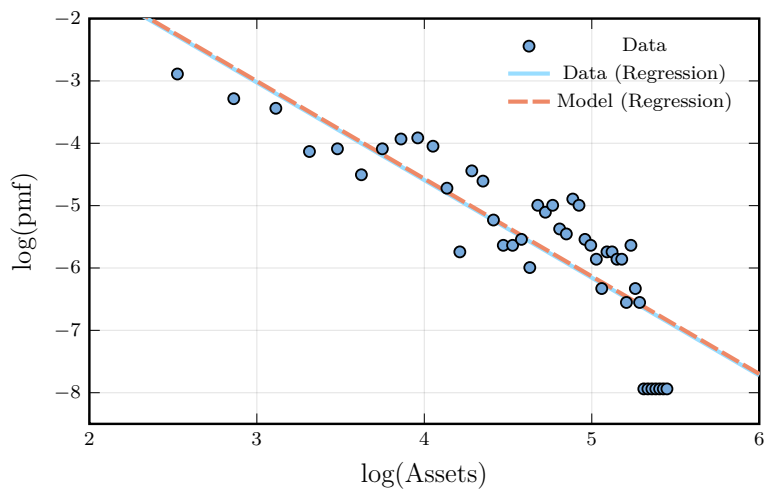
Source: European Central Bank, Statistical Data Warehouse (SDW). *Note*: The composition is expressed as percentages of total assets. Model counterparts in steady-state. Data corresponds to the consolidated balance sheet of euro area Monetary Financial Institutions (MFIs), excluding the Eurosystem, reported by the European Central Bank. *Loans* include loans to the private sector, to the general government, and other risky assets. *ST securities and reserves* include short-term securities holdings, operations with national central banks (repos and securities lending), and other short-term external assets. *Deposits* include retail deposits of different maturities, external and other liabilities. *Wholesale funding* corresponds to debt securities issued. *Equity capital* comprises capital and reserves.

[14] Appendix B.4 provides additional analysis of this categorization. In particular, we show that these patterns have remained stable and extend across loan categories, affecting both loans to households and to non-financial corporates.

[15] MFIs includes deposit-taking institutions (banks) and money market funds (MMFs). We cannot distinguish MMF from the deposit-taking institutions for the entire time series, but the presence of MMFs is immaterial given their size. For example, in 2024Q4, MMFs assets totaled €1.9 trillion, or less than 5% of deposit-taking institutions' assets (€38.6 trillion). Appendix B.1 details the composition of MFIs and the time series used.

Asset-size distribution. The model generates a steady-state distribution of bank assets that closely mirrors the asset distribution observed for euro area banks.[16] Figure 2 compares the model's and the empirical right tails of the asset distribution in the log-log space, illustrating the substantial heterogeneity in bank sizes produced by the model. The model reproduces the power-law behavior observed in the data. This emerges endogenously from the combination of size-independent growth rates and stochastic exit, consistent with Gabaix (2009).

Figure 2: Bank asset size distribution: Tail behavior.



Note: The blue dots represent equally spaced bins in the right tail of the empirical distribution of banks' assets. *Data (Regression)*: We fit a power law distribution of the form $f(x) = \bar{A}x^{\psi}$, where ψ captures the tail behavior of the asset distribution and denotes the slope when regressing the log-pdf on the log of these bins. The light blue line is the predicted relationship from such a regression. *Model (Regression)*: The dashed red line is the model-equivalent to the light blue line based on the steady-state distribution. To make the model and data distributions comparable, we have scaled the asset values so their means match in both the model and the data.

Capital ratio distribution. Table 3 reports the distribution of capital ratios in the data and in the model.[17]

[16] We characterize the distribution of banks' assets from an unbalanced bank-level panel from *S&P* Global, a proprietary source. This quarterly dataset covers a sample of 70+ euro area banks for the period from 2013 to 2020 and includes information on common equity tier 1 (CET1) capital levels, risk-weighted assets, and total assets. See Appendix B.3 for more details.

[17] Since the gradual implementation of Basel III in 2013, capital ratios for euro area banks have increased steadily. This adds additional dispersion to the distribution. To adjust the empirical distribution for time trends, we normalize each period by subtracting its mean and then re-centering the data using the mean capital ratio in 2019.

The model distribution captures its empirical counterpart well. The capital ratio of banks in the first percentile of the distribution is 9.7 percent in the model and 9.4 in the data, slightly above the regulatory limit of 7 percent. Both in the model and in the data, a substantial part of the mass is close to the constraint imposed by capital regulation: banks try to operate above the constraint to avoid being liquidated. The capital ratio of banks in the 40th percentile is 12.7 in the model and 13.5 in the data. For those banks in the upper half of the distribution, the average capital ratio is 13.1 in the model and 18.7 in the data, reflecting the fact that the model underestimates the mass of banks with high leverage levels.

To understand this, we also compare the model with the distribution of capital in large banks, defined as those with assets exceeding 10 billion euros, in the second column of Table 3. The model's fit improves for the entire capital distribution. The average for the mass of banks in the upper half, for instance, is now 14.7 in the data (compared with 13.1 in the model). We attribute this fact, among other things, to additional regulatory constraints that our model is not able to capture. In particular, banks should satisfy a Minimum Requirement for Own Funds and Eligible Liabilities (MREL).[18] While medium and large banks typically issue contingent liabilities to satisfy this requirement, small local banks typically satisfy this constraint by issuing additional CET1 capital, which may explain the CET1 levels well above 15%.

In any case, the underestimation of the right tail is inconsequential for our results. As we show next, the behavior of banks far away from the constraint is pretty much homogeneous. Therefore, aggregate dynamics are affected by the left tail and not by the right one.

3.4 Time-series moment fit

Responses to a monetary tightening. The main focus of interest is on how heterogeneous banks respond to monetary policy shocks, along various dimensions. To evaluate the model's ability to account for the bank lending channel, we compare the model-generated impulse response functions (IRFs) with their empirical counterparts. All the empirical IRFs are estimated using a local projections approach (Jordà, 2005; Jordà et al.,

[18]This is a regulatory framework that requires financial institutions to hold a sufficient amount of own funds and eligible liabilities to absorb losses and potentially recapitalize in case of failure.

Table 3: Capital-ratio distribution

	All Banks	Large Banks	Model
1st Percentile	9.36	9.68	9.71
5th Percentile	11.22	10.91	11.10
10th Percentile	11.68	11.28	11.66
20th Percentile	12.31	11.67	12.18
30th Percentile	13.03	12.00	12.46
40th Percentile	13.54	12.41	12.65
Avg. Top 50%	18.71	14.73	13.13

Sources: S&P Global and ESRB supervisory data on European banks' capital requirements. Capital ratios are defined as CET1 capital over risk-weighted assets. The sample corresponds to 60+ large and medium-sized euro area banks from 2013 to 2020. Medium-sized banks are defined as those with assets exceeding 10 billion euros, while those with assets exceeding 100 billion euros are considered large.

2015).[19]

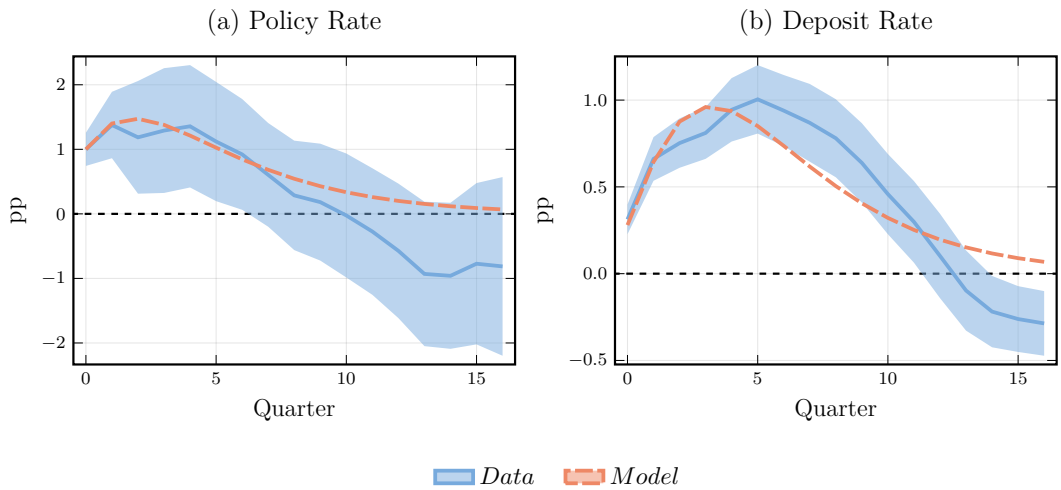
To obtain the model's response of new loans to an unexpected increase in the policy rate r^M, we compute the transitional dynamics after an unanticipated (MIT) shock following an algorithm similar to Boppart et al. (2018).[20] The loan rates and quantities in the model are determined in equilibrium via market-clearing. As explained above, the deposit side to induce similar deposit-rate paths as observed in the empirical IRF. Thus, for new loans, quantities and prices are jointly determined by demand and supply. For deposits, rates are de facto exogenous with quantities determined by the demand for funding by banks. All in all, the model takes as exogenous inputs the projected paths for both the policy rate and the deposit rate after a 1 percentage point increase in the policy rate, r^M, estimated from the data.

Figure 3 presents the IRFs of the exogenous rate processes. Panel (a) displays the estimated trajectory of the policy rate, and Panel (b) shows the corresponding path for

[19]We estimate empirical impulse responses using a balanced panel comprising the ten largest euro area countries, which allows us to construct a dataset without data gaps. We classify a country as operating in a VR if its share of variable-rate lending is above 50%, and as FR otherwise. The VR group includes Spain, Portugal, Italy, Finland, Ireland, and Austria, while the FR group consists of Germany, France, Belgium, and the Netherlands. Appendix B.5 provides details on the local projection estimation. Appendix B.6 shows that extending the panel to include all twenty euro area countries yields similar results.

[20]This is equivalent to solving a model with aggregate risk by a first-order perturbation method.

Figure 3: Targeted impulse responses



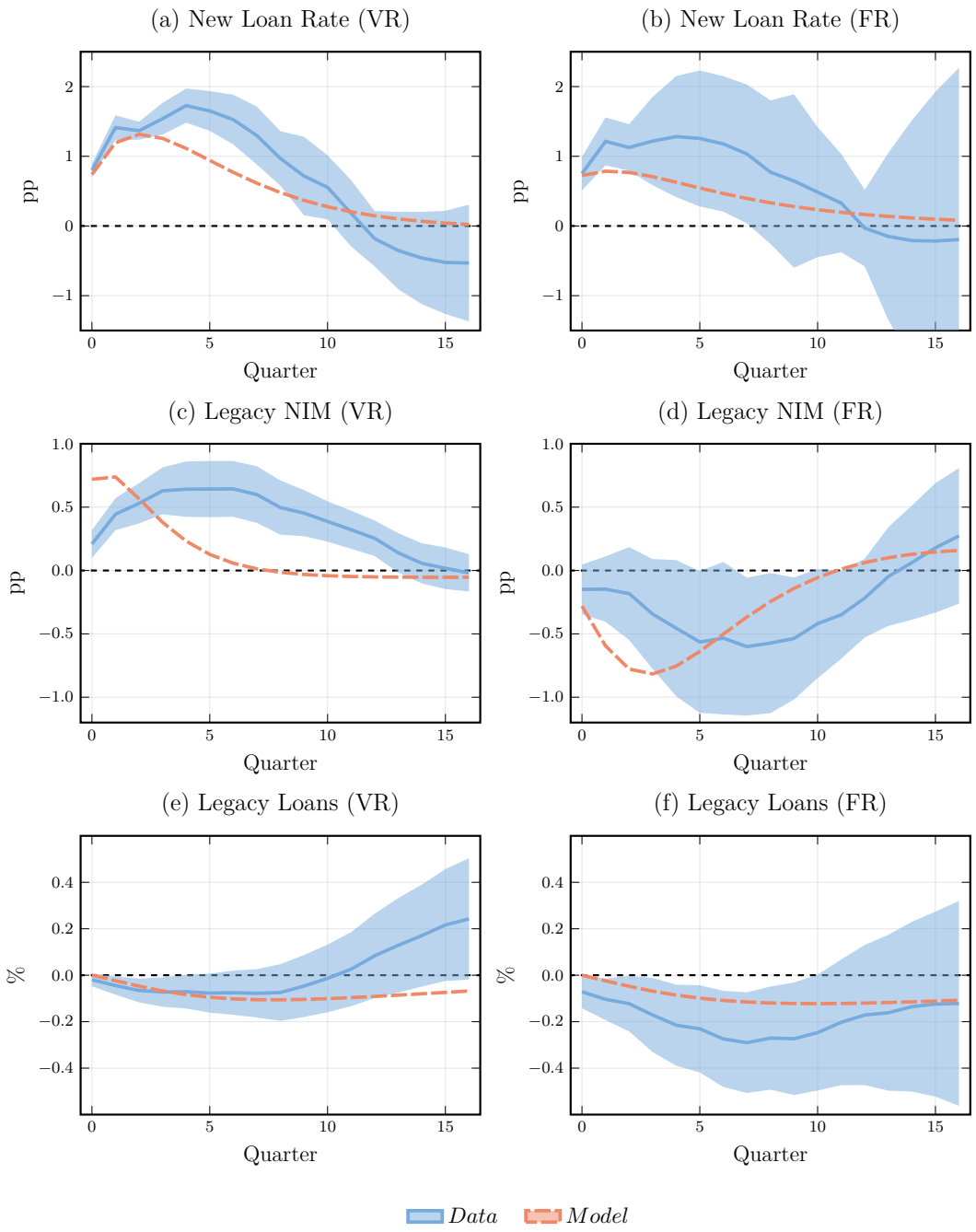
Note: Solid blue lines show the empirical impulse responses to a monetary policy shock and dashed red lines compute the model counterparts. Light blue bands show the 95% confidence intervals. Panels (a) and (b) report the responses of the policy rate and the deposit rate, respectively. See Appendix B.5 for details.

the deposit rate. The solid blue line and the blue shaded areas correspond to the point estimates and confidence bands of the IRFs. The dashed lines in those top panels are the exogenous rate paths induced into the model. The figure shows how the shocks are calibrated to roughly match the dynamics in the data.

We further analyze the IRFs of variables that are not calibration targets in Figure 4 . This allows us to assess the model's fit along other bank-balance sheet variables. The panels on the left represent VR countries, whereas the panels on the right represent FR counterparts. The fit to the data is good in the cases of the rate charged on new loans and of the amount of legacy loans. The model captures the fact that the passthrough to new loan rates is higher in VR economies than in FR ones (panels a and b). The model also captures how the decline in the volume of legacy loans is larger and more persistent in FR economies (panels e and f). Regarding the NIM (panels c and d), the response of the NIM is positive for VR countries and negative for FR ones both in data and model, a feature that we explain in the next section. The model IRFs of NIM are in the order of magnitude of those in the data, but miss the dynamics, which are more persistent in the data.[21]

[21]The NIM for legacy loans is the difference between average interest rate on the stock of legacy loans and the average deposit rate.

Figure 4: Untargeted impulse responses

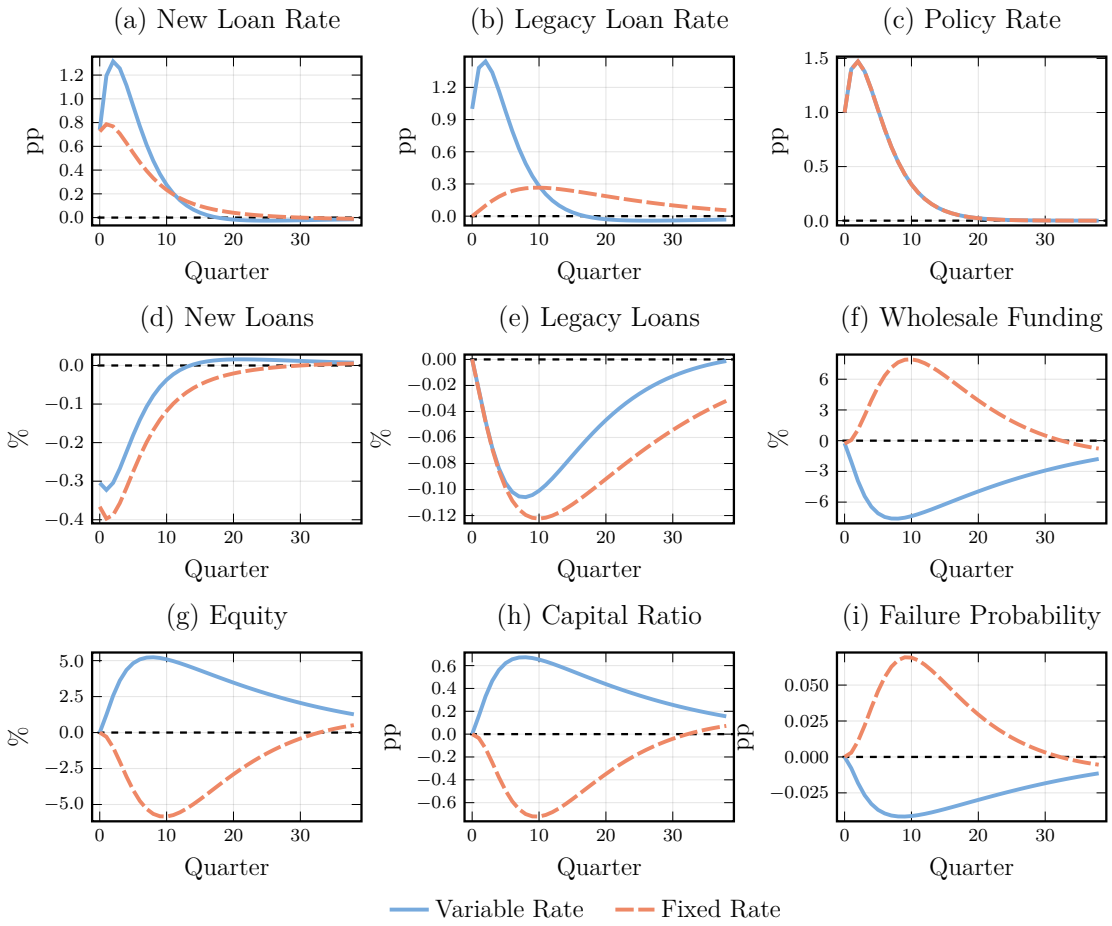


Note: Solid blue lines show the empirical impulse responses to a monetary policy shock and dashed red lines compute the model counterparts. Light blue bands show the 95% confidence intervals. Panels (a) and (b) report the response of the interest rate on new loans, panels (c) and (d) report the response of the legacy NIM, panels (e) and (f) report the response of legacy loans, distinguishing between FR and VR countries in the data and in the model. See Appendix B.5 for details.

4. Inspecting the mechanism

In this section, we use the model to analyze how ex-ante and ex-post heterogeneity impact the bank-lending channel. We first consider the role of ex-ante heterogeneity by comparing the IRFs in FR and VR economies—to a one percentage point increase in the policy rate (the remuneration rate of central bank reserves). Then, we evaluate the role of ex-post heterogeneity by looking at the IRFs of different banks in different points of the capital-ratio distribution. Finally, we contrast the model with a version with lower idiosyncratic credit risk.

Figure 5: Aggregate impulse response functions



Note: Impulse responses to a 1 p.p. increase in the policy rate. Solid blue lines show the economy with variable interest rate loans and the dashed red lines the economy with fixed interest rate loans.

Ex-ante heterogeneity. Figure 5 contrasts the aggregate-level IRFs to a one percentage point contractionary monetary policy shock for VR and FR economies. Following the same path of a policy shock (Panel c), banks' funding costs rise as the interest rate on wholesale debt increases in tandem. In contrast, the pass-through to deposit rates remains gradual, by construction (recall the explanation Figure 3).

This asymmetric response causes the marginal cost of funding to increase sharply, while the average funding cost remains more contained. The average funding cost is mainly driven by the more gradual increase in deposit rates, due to deposits representing the lion's share of banks' liability mix.

The increase in funding costs depresses new-loan origination (panel d). The contraction in the aggregate credit supply, in turn, leads to an increase in rates as entrepreneurs now compete for a smaller pool of new loans.

For VR banks, the initial pressure on funding costs is offset by a rapid pass-through of the policy rate to both new and legacy loan rates (solid blue line, Panels a and b). This widens the NIM (as seen in Figure 4), translating into higher profitability, which in turn increases their equity and capital ratios (Panels g and h). In stark contrast, FR banks (dashed red line) experience a severe and prolonged compression in their NIM, as their funding costs rise while income from their fixed-rate legacy portfolio remains stagnant (panel b, the higher interest rate on new loans only increases the average rate in the banks' portfolio as new loans slowly replace maturing legacy ones).[22] These dynamics precipitate an erosion of their equity and capital ratios. The decline in equity is partially covered by an increase in the reliance on wholesale funding (Panel f). The asymmetry in the response of equity represents the main amplification channel explaining the diverging responses across the two types of banking sectors.

Ultimately, the deterioration in bank capitalization under a FR regime leads to a significantly sharper decline in new loan origination compared to the VR system (Panel d). The decline in capital forces FR banks to reduce their lending proportionally more, to avoid an excessive increase in leverage. The result is a more prolonged contraction in the loan volume under fixed rates (panel e), which explains the empirical evidence presented in Figure 4.

The mechanism highlighted in our model aligns with the findings of Hoffmann et al. (2018), who document that cross-sectional variation in European banks' interest rate risk exposures is primarily driven by cross-country differences on the asset side—arising

[22] Appendix B.6 shows that the empirical responses are not driven by a core-periphery grouping.

from loan-rate fixation conventions—rather than by heterogeneity on the liabilities side. They show show that net worth increases with interest rates for roughly half of the banks in their sample due to the different exposure in VR economies.

The monetary tightening has opposing effects on financial stability: the probability of bank failure increases markedly in the FR economy, owing to the decrease in banks' capitalization, whereas it declines for VR banks (Panel i). The heterogeneity in loan-rate fixation patterns is therefore a critical determinant of both the amplification of monetary policy and its consequences for financial stability.

All in all, we can conclude that ex-ante heterogeneity leads to a quantitative difference in the bank lending channel: the elasticity of new lending is about 1/3 larger in fixed-rate economies. Beyond this significant quantitative difference, the implications of bank capital, critical for financial stability, go in opposite directions. This is in line with previous empirical analysis, such as Altunok et al. (2023) who document that U.S. banks with a higher share of adjustable-rate mortgages in their overall loan portfolio benefit from rate hikes. When monetary policy tightens, these banks earn higher interest income, exhibit stronger stock price reactions, and expand credit supply, as opposed to banks with a larger share of fixed-rate mortgages in their loan portfolios.[23]

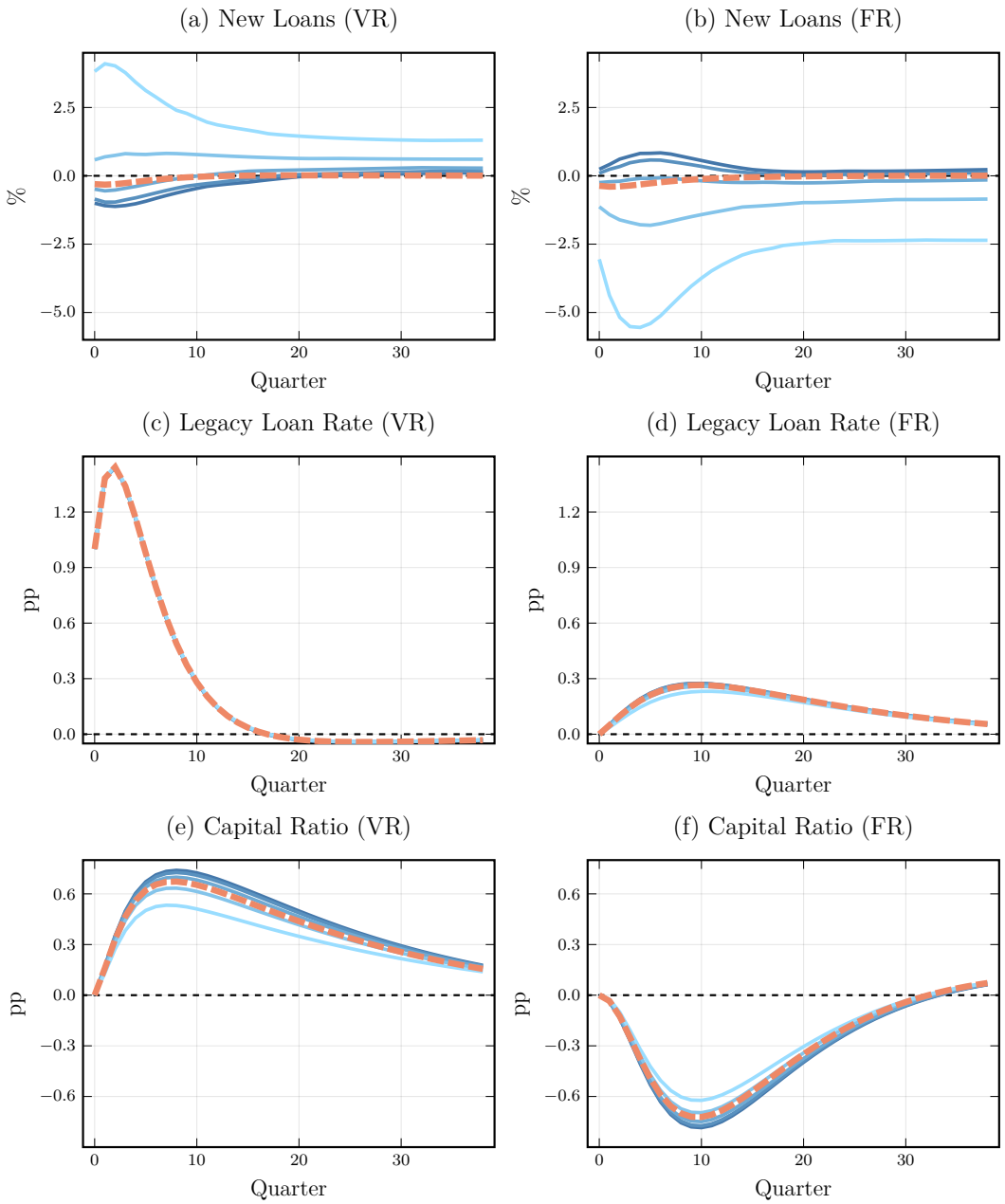
Ex-post heterogeneity. The aggregate responses mask a substantial degree of heterogeneity in the transmission of monetary policy to bank lending. To show this, we disaggregate the responses to explore the role of ex-post heterogeneity in bank leverage. Figure 6 presents the impulse responses for banks in different quantiles of the steady-state capital-ratio distribution, where lighter shades represent more highly leveraged institutions (lower capital ratios).

Consistent with the empirical literature (e.g, Jiménez et al., 2012; Dell'Ariccia et al., 2017; Altavilla et al., 2020), high-leverage banks (in light blue) are a key margin of transmission. In both FR and VR systems, high-leverage banks exhibit a stronger lending response to the monetary policy shock, though in opposite directions (panel a and b).

To understand this result, notice first that, as discussed above, a monetary contraction increases loan rates more in VR economies than in FR. The transmission is identical

[23] Our work also provides a clear mechanism for the asymmetry in monetary policy transmission that these authors document. During rate hikes, banks with predominant ARM portfolios expand credit supply much more strongly than FRM-banks as their net-interest income and capital position improve.

Figure 6: Individual impulse response functions



Note: The dashed red line shows the aggregate impulse response for the respective variable in fixed-rate (FR) and variable-rate (VR) banking systems. The solid blue lines show the impulse responses of banks that are at the 1st, 10th, 50th, 90th, or 99th percentile of the capital ratio distribution. The lightest shade of blue corresponds to the 1st percentile, i.e., the bank that is closest to the capital requirement in the steady state, while the other percentiles are shown with increasingly darker shades of blue.

across banks in a given economy, because they operate in a competitive market for loans (panels c and d). As deposit and policy rates are also identical, the impact on profitability before taking new loan decisions depends only on leverage, with more leveraged institutions experiencing higher gains or losses. This effect is dominated by the fact that, given a certain gain or loss, the banks with a lower the capital ratio—and thus closer to the regulatory requirement—have a much higher elasticity of new lending to changes in their profitability. This latter effect explains why low capital (high leverage) banks increase their loans in VR economies while reducing them in FR economies (panels a and b).

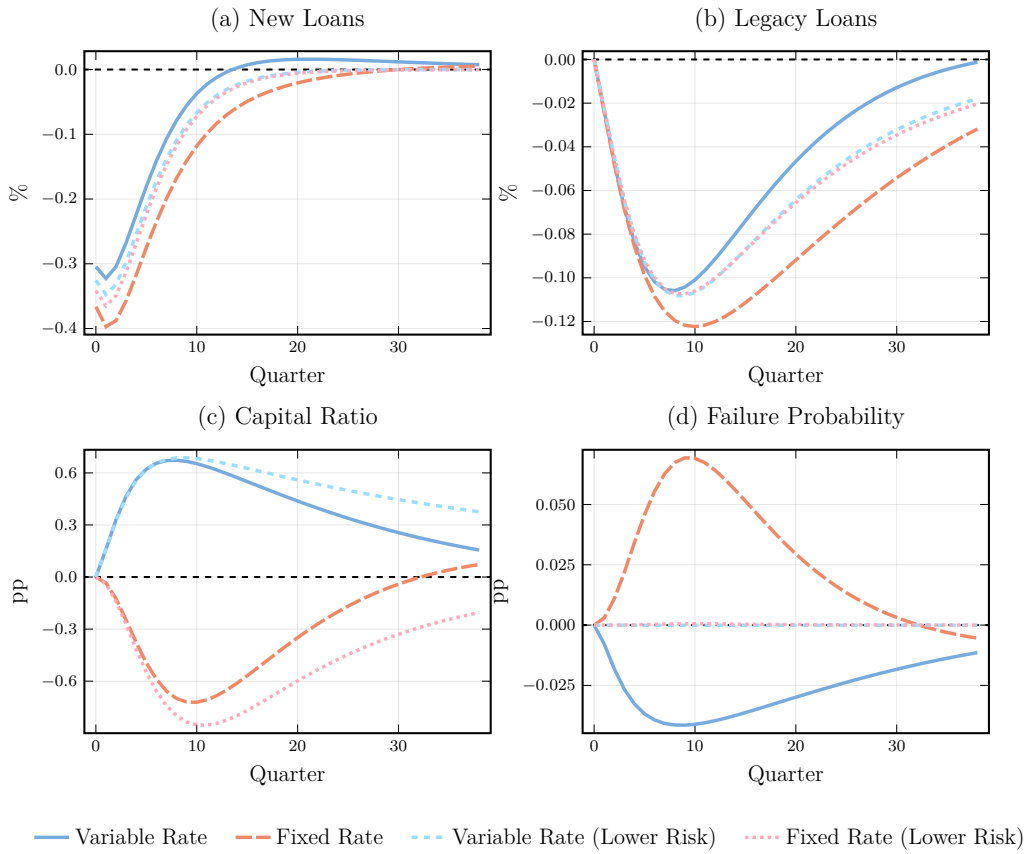
The proportionally higher increase in new loans for low-capital banks in VR economies trumps the proportionally higher increase in equity, the result is that the capital ratio after new loan decisions are taken increases proportionally less for low-capital banks. In the case of high-capital (low-leverage) banks, the effect is the opposite, as the increase in capital is larger than the increase in new loans, so that their capital ratio increases proportionally more than the aggregate. FR systems are the mirror image in this respect.

The bottom line is that there is a *credit redistribution* among banks in response to a monetary policy contraction. In VR economies high-leverage banks increase their share in overall credit whereas in FR system they reduce it.

Interactions between both types of heterogeneity. Figure 7 compares the responses of loan quantities and financial stability variables in both economies (FR and VR) against counterfactual cases where we significantly reduce the level of idiosyncratic risk (in dotted light lines).

This experiment provides a quantitative benchmark to the theoretical results of Proposition 1 above. The Proposition includes two conditions: (i) that capital requirements are not binding for all banks at all times; and (ii) that banks and entrepreneurs discount future cash flows at identical rates. In the current experiment Condition (i) is approximately satisfied, as the low idiosyncratic risk makes banks relatively similar and reduces the share of institutions at, or close to, the regulatory constraint. Condition (ii), however, is not satisfied as the discount rates of banks and entrepreneurs are different. Nonetheless, despite the violation of Condition (ii), credit dynamics are almost identical in FR and VR economies in this case, as highlighted in Panels a and b. This happens even though the policy rate shock moves their capital ratios in opposite directions (panel c). The failure probability remains null, reflecting the fact that no bank gets

Figure 7: Impulse response functions — Lower idiosyncratic risk



Note: The impulse responses denoted "Variable Rate" and "Fixed Rate" correspond to the baseline calibration. "Variable Rate (Lower Risk)" and "Fixed Rate (Lower Risk)" denote alternative parameterizations where we reduce ρ from 0.51 to 0.1, meaning that in these economies there is substantially lower idiosyncratic risk.

closer to the regulatory constraint.

This exercise quantitatively show one of the key result of this paper: that heterogeneity in interest-rate risk exposure only matters insofar as there is also ex-post leverage heterogeneity across banks in the presence of capital constraints.

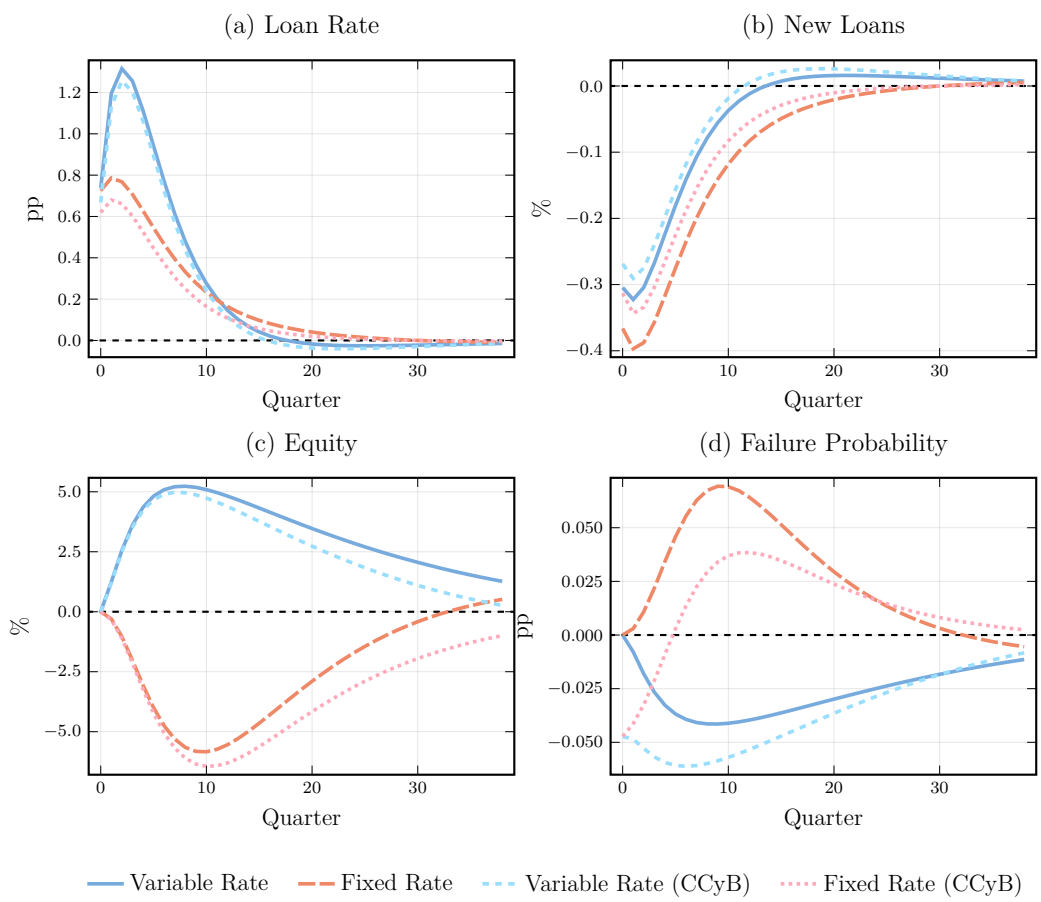
5. Monetary policy and financial stability

Financial stability and monetary policy conflicts. Our findings have important implications for the design and coordination of monetary and macroprudential policies. The most common macroprudential tool is the countercyclical capital buffer (CCyB). The CCyB represents an increase in capital requirements during credit expansions and

a tightening during credit contractions. The objective is to avoid excessive volatility in credit cycles that may threaten financial stability.

As monetary policy also affects credit, the natural question is how these two policies interact. Figure 8 reproduces the impulse response functions following a monetary policy tightening in a VR (solid blue line) and FR (dashed red line) economies. It also displays the same responses when the macroprudential authority releases the CCyB by 1pp. The CCyB release is modeled as a changed in capital requirement γ, where $\gamma_t - \bar{\gamma} = \rho_{ccyb}(\gamma_{t-1} - \bar{\gamma})$ and $\rho_{ccyb} = 0.95$. This implies that the capital requirement slowly reverts back to its steady-state value.

Figure 8: Impulse response functions — Interest rate increase + CCyB release



Note: The impulse responses denoted "Variable Rate" and "Fixed Rate" correspond to the baseline calibration. "Variable Rate (CCyB)" and "Fixed Rate (CCyB)" denote alternative parameterizations where γ is reduced by 1pp on impact of the policy rate increase and then gradually reverts to its steady-state value.

A temporary relaxation in capital requirements can dampen the differential effects

of monetary policy across banking systems. Looser capital requirements reduce the fraction of banks operating near regulatory constraints. The immediate effect is that the failure probability is reduced on impact, as banks have higher regulatory space (panel d).

Furthermore, a reduction in the capital requirement pushes the whole distribution of banks further away from the regulatory constraint. This has an asymmetric effect: as equity dynamics are the opposite in FR and VR economies (panel c), reducing the impact of equity changes on loan provision makes the response of new loans more similar in both types of economies (panel b). This is no surprise, as it relates back to Proposition 1: the less constrained banks are by capital requirements, the less relevant ex-ante heterogeneity matters.

The opposite result (not shown) is also true: if macroprudential policy tightens during a monetary contraction, the asymmetric impact on credit provision becomes acuter. This asymmetry makes the trade-offs facing monetary policymakers in a monetary union more acute, as macroprudential policy changes can inadvertently increase the divergence in regional responses to uniform policy.

Financial stability origins of monetary policy gradualism. Finally, we focus on how different paths of monetary policy tightening that produce similar cumulated increases in interest rates may have different financial stability implications in FR or VR banking systems.

To this end, we compare a range of different policy rate paths that all yield the same area under the policy rate IRF. This is inspired by the fact that in a standard 3-equation New Keynesian model, two alternative policy rate paths are equally stimulative for economic activity, meaning that they have the same impact on the output gap at $t = 0$, as long as the area under the ex-ante real rate IRF is the same under both policy rate paths.[24]

[24] To see this note the 3-equation New Keynesian model has an IS curve of the form

$$x_t = \mathbb{E}_t(x_{t+1}) - \frac{1}{\gamma}(i_t - \mathbb{E}_t(\pi_{t+1}) - r_t^n)$$

where x_t is the output gap, i_t is the nominal interest rate, π_t is the inflation rate, and r_t^n is the natural rate of interest. Note that all variables are expressed in log-linear deviations from steady state. Iterating on this equation and using the ex-ante real rate $\hat{r}_t^M = r_t^M - \tilde{r}_t^M = i_t - \mathbb{E}_t(\pi_{t+1})$, we can write the output gap as

$$x_t = -\frac{1}{\gamma}\mathbb{E}_t \sum_{j=0}^{\infty}\left(\hat{r}_{t+j}^M - r_{t+j}^n\right).$$

To formalize the comparison, we consider variations of the AR(2) process used in the baseline calibration

$$\hat{r}_t^M = \phi_1 \hat{r}_{t-1}^M + \phi_2 \hat{r}_{t-2}^M + \sigma \varepsilon_t \,.$$

In particular, we exploit the fact that the process can be rewritten as two first-order processes

$$\begin{aligned} \hat{r}_t^M &= \mu_1 \hat{r}_{t-1}^M + z_t \,, \\ z_t &= \mu_2 z_{t-1} + \sigma \varepsilon_t \,, \end{aligned}$$

where μ_1 and μ_2 are the roots of the characteristic polynomial $x^2 - \phi_1 x - \phi_2 = 0$.

One can show that in our perfect foresight environment, the area under the policy rate path is given by

$$\sum_{t=0}^{\infty} \hat{r}_t^M = \frac{\sigma}{(1-\mu_1)(1-\mu_2)} \,.$$

Therefore, if we fix μ_1 at the value in the baseline calibration, more gradual policy rate paths that are as stimulative as the baseline path are characterized by a higher value for μ_2 and lower values for σ such that the area under the policy rate path remains the same.

Let's consider what the impact of a more gradual monetary policy tightening is in a FR economy. As the path for rates becomes more gradual, the initial rate hike becomes smaller. This reduces the initial negative effect on the NIM (panel b in Figure 9), at the expense of making the decline in profitability more persistent. This softens the decline in equity as well as in new loans (panel d).

We turn next to the VR economy. In that case, the lower initial hike now reduces the positive effect on NIM (panel a), which amplifies the decline in new loans (panel e), the opposite effect to FR systems. Furthermore, the prolonged period of high rates depressed more demand in this case, as entrepreneurs forecast the whole path of rates.

Failure probabilities reflect these developments: the sharp distinction between FR and VR economies progressively blurs as the tightening becomes more gradual.

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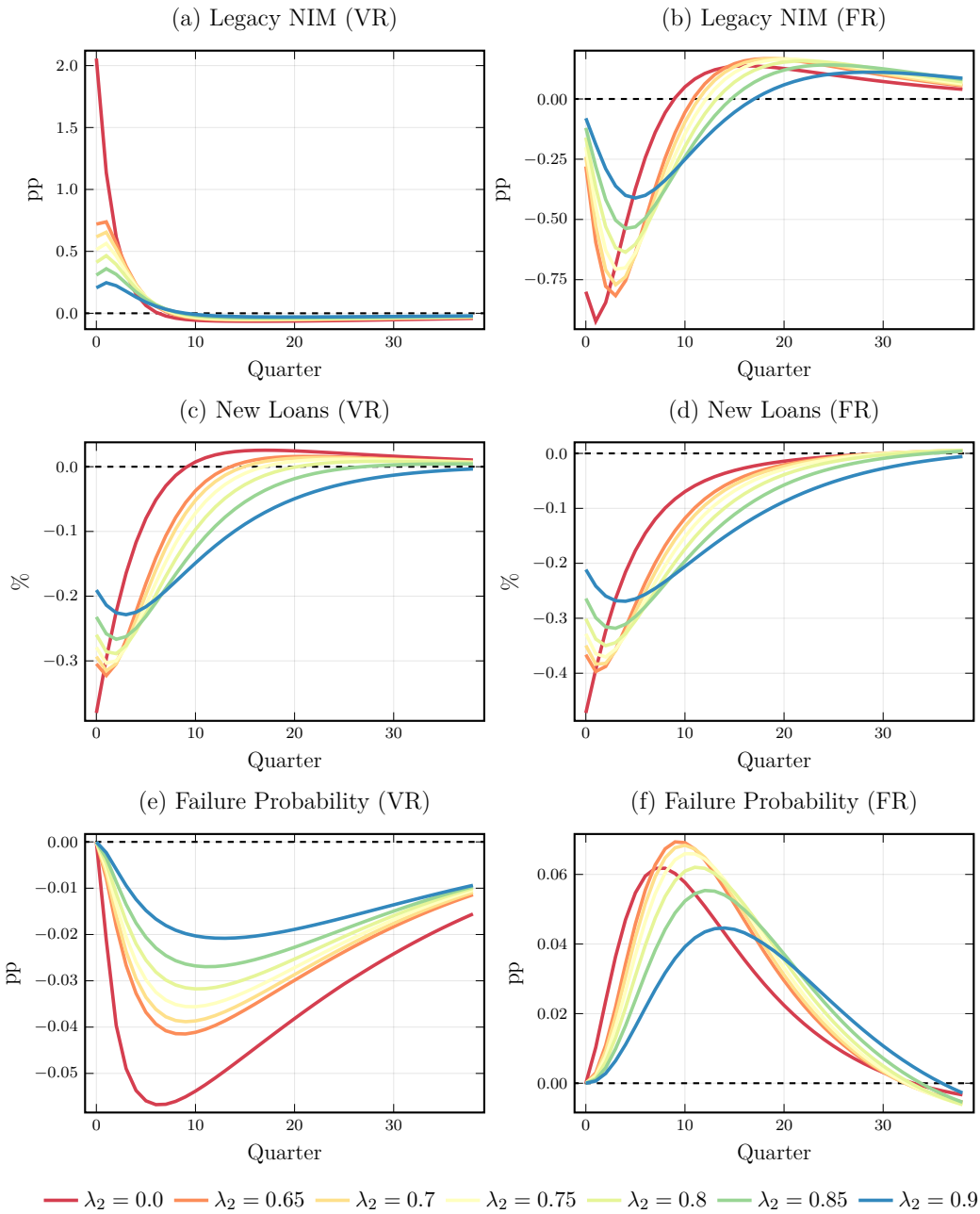
meaning that the output gap is the sum of the expected deviations of the ex-ante real rate and the natural rate of interest. Thus, two alternative policy paths such that

$$\mathbb{E}_t \sum_{j=0}^{\infty} \hat{r}_{1,t+j}^M = \mathbb{E}_t \sum_{j=0}^{\infty} \hat{r}_{2,t+j}^M$$

yield the same output gap, i.e., have the same impact on the economy, for a given natural rate path.

Figure 9: Effects of Gradualism



Note: Panels (a) and (b) show the response of the legacy NIM, panels (c) and (d) show the response of new loans, and panels (e) and (f) show the failure probability, distinguishing in each case between VR and FR economies. The colors from red to blue correspond to different degrees of gradualism as captured by λ_2, where $\lambda_2 \in \{0.0, 0.65, 0.7, 0.75, 0.85, 0.9\}$, i.e., red corresponds to an AR(1) process, while blue is the most gradual AR(2) process.

Our analysis provides a theoretical foundation for gradualism in monetary policy conduct. Rapid policy rate changes force abrupt deleveraging in FR banking systems that amplify the contractionary effects. In contrast, gradual and well-communicated policy adjustments allow banks to rebuild capital buffers and adjust their portfolios more smoothly, reducing the severity of the lending contraction while achieving the same cumulative policy stance.

The insights in this section suggest that effective macroeconomic stabilization requires careful coordination between monetary and prudential authorities, with particular attention to the timing and sequencing of policy interventions across these domains.

6. Conclusion

This paper investigates how bank heterogeneity shapes the transmission of monetary policy.

Our central finding is that the ex-ante heterogeneity between loan-pricing regimes is quantitatively important for the bank lending channel. In a FR economy, banks are unable to reprice legacy assets to offset rising funding costs after a monetary tightening, reducing their profitability and their capital positions. This forces a sharper credit contraction relative to VR economies, in which loan repricing combined with a gradual pass-through to deposit rates makes the effect of a monetary tightening positive for profitability. Furthermore, the shock also raises the probability of bank failure in FR economies, highlighting the financial stability risks.

Furthermore, the model demonstrates that banks' leverage acts as a powerful amplifier, but its effects are contingent on the loan-pricing regime. Consistent with the literature, high-leverage banks are the primary margin of transmission. However, in a FR economy, their proximity to regulatory constraints forces them to cut lending most aggressively in response to capital depletion. Conversely, in a VR economy where monetary tightening boosts profitability, these same high-leverage banks expand their loan origination most. Our analysis thus reveals that the interplay between loan pricing and bank capitalization is fundamental to understanding both the aggregate strength of the transmission and the heterogeneous impact of monetary policy.

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Appendices

A. Model derivations

A.1 Portfolio credit risk

It is assumed that individual banks face limits in fully diversifying their loan portfolio and that loan defaults in the portfolio of bank j are correlated according to the *single risk factor* model of Vasicek (2002), in which the failure of the loan i from bank j is driven by the realization of a latent random variable:

$$\xi_{ijt+1} = -\Phi^{-1}(p) + \sqrt{\rho} z_{jt+1} + \sqrt{1-\rho}\varepsilon_{it+1}, \tag{20}$$

where $\Phi(\cdot)$ denotes the cdf of a standard normal random variable and $\Phi^{-1}(\cdot)$ its inverse, z_{jt+1} is a bank-idiosyncratic risk factor that affects all projects in bank's j portfolio, ε_{it+1} is a project-idiosyncratic risk factor that only affects the loan i, and $\rho \in [0,1]$ determines the extent of correlation in loan failures. It is assumed that z_{jt+1} and ε_{it+1} are standard normal random variables, independently distributed from each other, as well as across time, banks, and loans.

The loan i fails when $\xi_{ijt+1} < 0$. The deterministic term $-\Phi^{-1}(p)$ in (20) ensures that the unconditional probability of failure of project i satisfies:

$$\Pr(\xi_{ijt+1} < 0) = \Pr\left[\sqrt{\rho} z_{jt+1} + \sqrt{1-\rho}\varepsilon_{it+1} < \Phi^{-1}(p)\right] = \Phi\left[\Phi^{-1}(p)\right] = p. \tag{21}$$

Notice that for $\rho = 0$ the bank-idiosyncratic risk factor does not play any role and loan failures are statistically independent, while for $\rho = 1$ the entrepreneur-idiosyncratic risk factor does not play any role and loan failures are perfectly correlated within each bank. By the law of large numbers, the failure rate ω_{jt+1} (the fraction of loans within a bank's portfolio that fail) for a given realization of the bank-idiosyncratic risk factor z_{jt+1} coincides with the probability of failure of a (representative) project i conditional on z_{jt+1}; that is,

$$\begin{aligned}\omega_{jt+1} =& \xi(z_{jt+1}) = \Pr\left(-\Phi^{-1}(p) + \sqrt{\rho} z_{jt+1} + \sqrt{1-\rho}\varepsilon_{it+1} < 0 | z_{jt+1}\right) \\ =& \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho} z_{jt+1}}{\sqrt{1-\rho}}\right).\end{aligned} \tag{22}$$

From here it follows that the CDF of the loans' failure rate is

$$\begin{aligned} F(\omega_{jt+1}) =& \Pr\left[\xi(z_{jt+1}) \leq \omega_{jt+1}\right] = \Pr\left[z_{jt+1} \geq \xi^{-1}(\omega_{jt+1})\right] \\ =& \Phi\left(\frac{\sqrt{1-\rho}\Phi^{-1}(\omega_{jt+1}) - \Phi^{-1}(p)}{\sqrt{\rho}}\right). \end{aligned} \tag{23}$$

A.2 Conditions for risk-free wholesale debt

The balance sheet of the bank, after substituting for the binding constraints (4) and (8), reads:

$$L_{jt} + N_{jt} + \theta\alpha L_{jt} = \alpha L_{jt} + (1-\theta)B_{jt} + E_{jt}.$$

Solving for B_t:

$$B_{jt} = \frac{1}{1-\theta}\left([1+\alpha(\theta-1)]L_{jt} + N_{jt} - E_{jt}\right).$$

Consider the worst possible realization for the iid shock ($\omega_{jt+1} = 1$). If wholesale debt is collateralized, debt holders recover at most:

$$(1-\mu)\left[(1+r_t^M)M_{jt} + (1-\lambda)(L_{jt}+N_{jt})\right],$$

where μ is a proportional bank resolution cost. For debt to be risk free, we need:

$$(1+r_t^B)B_{jt} \leq (1-\mu)\left[(1+r_t^M)M_{jt} + (1-\lambda)(L_{jt}+N_{jt})\right].$$

Using B_{jt} and the equilibrium condition $r_t^B = r_t^M$, this can be rewritten as:

$$(1+r_t^B)\frac{1-(1-\mu)\theta}{1-\theta}\left([1+\alpha(\theta-1)]l_{jt} + n_{jt} - 1\right) \leq (1-\mu)\left[(1+r_t^M)\theta\alpha l_{jt} + (1-\lambda)(l_{jt}+n_{jt})\right]$$

where everything has been expressed in ratios to equity. We numerically confirm this condition to be satisfied across the state space for our calibration.

A.3 A microfundation for aggregate deposits demand

This section provides a microfoundation for the deposit supply function used in the main text. We develop a household problem that generates demand functions for

deposits ¿and monetary holdings by banks?, which can then be aggregated to obtain the supply of deposits to banks. This micro-foundation is critical to allow us to treat r_t^M and r_t^D as exogenous paths with perfectly-elastic supply schedules.

The household problem. Consider a representative household that derives utility from two consumption goods, C_t and C_t^H, and from holding a bundle of assets. The household solves the following recursive problem:

$$V_t^H(A_{t-1}^H, D_{t-1}^H) = C_t^H + U\left(C_t + \ell(A_{t-1}^H, D_{t-1}^H)\right) + \beta V_{t+1}^H(A_t^H, D_t^H), \tag{24}$$

subject to the budget constraint:

$$C_t + C_t^H + \underbrace{B_t^H + M_t^H}_{=A_t^H} + D_t^H = (1 + r_{t-1}^B)B_{t-1}^H + (1 + r_{t-1}^M)M_{t-1}^H + (1 + r_{t-1}^D)D_{t-1}^H. \tag{25}$$

The function $\ell(m, d)$ captures the liquidity services provided by the household's portfolio of assets. We assume a Cobb-Douglas aggregator:

$$\ell(A, D) = \kappa \frac{\left(A^\nu D^{1-\nu}\right)^{1-\vartheta}}{1-\vartheta}, \tag{26}$$

with $\nu \in (0, 1)$ and $\kappa, \vartheta > 0$. The parameter ν governs the relative importance of bonds versus deposits in providing liquidity services.

Solution. To solve the problem, we substitute the budget constraint into the objective function:

$$\begin{aligned} V_t^H(A_{t-1}^H, D_{t-1}^H) = {} & U\left(C_t + \ell(A_{t-1}^H, D_{t-1}^H)\right) - \left(C_t + A_t^H + D_t^H\right) \\ & + (1 + r_{t-1}^M)A_{t-1}^H + (1 + r_{t-1}^D)D_{t-1}^H + \beta V_{t+1}^H(A_t^H, D_t^H). \end{aligned} \tag{27}$$

We conjecture that $V_t^H(A_{t-1}^H, D_{t-1}^H)$ is linear in its arguments. Under this conjecture, we derive the first-order conditions.

The first-order condition with respect to C_t yields:

$$U'\left(C_t + \ell(A_{t-1}^H, D_{t-1}^H)\right) = 1. \tag{28}$$

The first-order condition with respect to A_t^H is:

$$-1 + \beta V_{A,t+1}^H(A_t^H, D_t^H) = 0, \tag{29}$$

where $V_{A,t+1}^H(A, D) \equiv \frac{\partial V_{t+1}^H}{\partial A}$. Using the envelope theorem and the fact that, in equilibrium, $r_t^M = r_t^B$ for all t:

$$V_{A,t+1}^H(A_t^H, D_t^H) = (1 + r_t^M) + U'\left(C_{t+1} + \ell(A_t^H, D_t^H)\right) \cdot \ell_A(A_t^H, D_t^H). \tag{30}$$

Substituting (28) evaluated at $t+1$, we have $U\left(C_{t+1} + \ell(A_t^H, D_t^H)\right) = 1$. Thus:

$$\beta\left[(1 + r_t^M) + \ell_A(A_t^H, D_t^H)\right] = 1. \tag{31}$$

Rearranging and using (28):

$$\ell_A(A_t^H, D_t^H) = \frac{1}{\beta} - (1 + r_t^M) := s_t^M, \tag{32}$$

where s_t^M denotes the spread between the household's rate of time preference and the return on bonds.

Proceeding analogously for deposits, we obtain:

$$\ell_D(A_t^H, D_t^H) = \frac{1}{\beta} - (1 + r_t^D) := s_t^D, \tag{33}$$

where s_t^D denotes the corresponding spread for deposits.

Given the functional forms in (26), we compute the partial derivatives:

$$\ell_A(A_t^H, D_t^H) = \frac{\nu\kappa\left[\left(A_t^H\right)^\nu \left(D_t^H\right)^{1-\nu}\right]^{1-\vartheta}}{A_t^H}, \tag{34}$$

$$\ell_D(A_t^H, D_t^H) = \frac{(1-\nu)\kappa\left[\left(A_t^H\right)^\nu \left(D_t^H\right)^{1-\nu}\right]^{1-\vartheta}}{D_t^H}, \tag{35}$$

Dividing (34) by (35) and using the first-order conditions (32) and (33):

$$\frac{\nu}{1-\nu}\frac{D_t^H}{A_t^H} = \frac{s_t^M}{s_t^D}. \tag{36}$$

This expression determines the optimal ratio of deposits to bonds as a function of the spreads.

Optimal quantities. To solve for the individual quantities, substitute the portfolio ratio back into the first-order conditions. From (32):

$$\frac{\nu\kappa\left[\left(A_t^H\right)^{\nu}\left(D_t^H\right)^{1-\nu}\right]^{1-\vartheta}}{A_t^H} = s_t^M, \tag{37}$$

which can be rewritten as:

$$\nu\kappa\left[\left(A_t^H\right)^{\nu}\left(D_t^H\right)^{1-\nu}\right]^{1-\vartheta} = s_t^M A_t^H, \tag{38}$$

From (36), we have:

$$D_t^H = \frac{(1-\nu)s_t^M}{\nu s_t^D} A_t^H. \tag{39}$$

Substituting into (38), we obtain the demand for bonds:

$$A_t^H = \left(\frac{\nu\kappa\left[\frac{(1-\nu)s_t^M}{\nu s_t^D}\right]^{(1-\nu)(1-\vartheta)}}{s_t^M}\right)^{\frac{1}{\vartheta}}. \tag{40}$$

Similarly, the demand for deposits satisfies:

$$D_t^H = \left(\frac{(1-\nu)\kappa\left[\frac{\nu s_t^D}{(1-\nu)s_t^M}\right]^{\nu(1-\vartheta)}}{s_t^D}\right)^{\frac{1}{\vartheta}}. \tag{41}$$

Market clearing. On the supply side, banks demand reserves according to the liquidity requirement:

$$M_t = \theta(D_t + B_t), \tag{42}$$

where D_t denotes the deposits supplied by banks and B_t denotes wholesale debt.

Market clearing in the deposit market requires:

$$D_t^H = D_t + D_t^S \tag{43}$$

where D_t^H denotes household demand for deposits and D_t^S the central banks' deposit supply.

Market clearing in the reserve market requires:

$$M_t^S = M_t^H + M_t, \tag{44}$$

where M_t^S denotes the supply of reserves by the central bank, M_t^H is household demand for bonds, and M_t is bank demand for reserves.

Key: The key is that the CB has two instruments, $\{D_t^S, M_t^S\}$, to target two rates: r_t^M and r_t^D. De facto, this makes the banks' deposit supply schedule perfectly elastic: any increase in their desired demand for deposits is offset by the central bank's position.

A.4 Derivation of Resource Constraint

Red text needs to be updated once the household problem is ready.

Let $y_t(l,x)$ denote the policy for variable y_t of a bank with leverage l and average loan rate/spread x on legacy loans and let $H_t(l,x,e)$ denote the joint distribution of leverage, the loan rate/spread on legacy loans and equity.[25] Furthermore, let $\bar{v}_{t+1}(l,x,\omega)$ denote the recovery value (per unity of equity) of a bank failing after a bad realization of ω at $t+1$. The recovery value can be written as:

$$\begin{aligned}
\bar{v}_{t+1}(l,x,\omega) &= (1-\mu)\Big((1-\omega)\left[(1+r_t^L(x))l + (1+r_t^N)n_t(l,x)\right] + \omega(1-\lambda)(l+n_t(l,x)) \\
&\quad + (1+r_t^M)m_t(l,x) - f\left(\frac{n_t(l,x)}{l}\right)l - \overline{\pi}\Big) \\
&= (1-\mu)\left(1+\pi_{t+1}(l,x,\omega) + (1+r_t^D)d_t(l,x) + (1+r_t^B)b_t(l,x)\right) \\
&= (1-\mu)\left(g_{t+1}(l,x,\omega) + \tau\pi_{t+1}(l,x,\omega) + (1+r_t^D)d_t(l,x) + (1+r_t^B)b_t(l,x)\right)
\end{aligned}$$

where

$$\begin{aligned}
\pi_{t+1}(l,x,\omega) &= (1-\omega)(r_t^L(x)l + r_t^N n_t(l,x)) + r_t^M m_t(l,x) \\
&\quad - r_t^D d_t(l,x) - r_t^B b_t(l,x) - \lambda\omega(l+n_t(l,x)) - f\left(\frac{n_t(l,x)}{l}\right)l - \overline{\pi}, \\
g_{t+1}(l,x,\omega) &= 1 + (1-\tau)\pi_{t+1}(l,x,\omega)
\end{aligned}$$

[25] For simplicity, we omit the subscript j in the derivations in this section.

are profits and the gross equity growth rate, respectively.

We start the derivation by combining the household budget constraint and the consolidated government budget constraint:

$$C_t^D + C_t^H + D_t + B_t^H + \Xi_t = \left(1 + r_{t-1}^D\right) D_{t-1} + \left(1 + r_{t-1}^B\right) B_{t-1}^H + \Pi_t^E + X_t - T_t \qquad \text{(HHs BC)}$$

$$T_t = \left(1 + r_{t-1}^B\right)\left(B_{t-1} + B_{t-1}^H\right) + \Upsilon_t - \left(B_t + B_t^H\right) - \tau \Pi_t, \qquad \text{(Gov BC)}$$

where

$$\begin{aligned}
\Upsilon_t &= \int \int_{\bar{\omega}}^{1} \left[\left(1 + r_{t-1}^D\right) d_{t-1}(l,x) + \left(1 + r_{t-1}^B\right) b_{t-1}(l,x)\right. \\
&\quad \left. - \bar{v}_t(l,x,\omega)\right] e \, \mathrm{d}F(\omega) \mathrm{d}H_{t-1}(l,x,e), && \text{(DIS \& Bank Resolution)} \\
\Xi_t &= \bar{\mathscr{F}}_{t-1} \bar{E}_t - \chi \int \int_0^{\bar{\omega}} g_t(l,x,\omega) e \, \mathrm{d}F(\omega) \mathrm{d}H_{t-1}(l,x,e), && \text{(Net Equity Injections)} \\
\Pi_t^E &= (1-p)\left((A - \bar{r}_{t-1}^{L*}) L_{t-1} + (A - r_{t-1}^N) N_{t-1}\right), && \text{(Entrepreneurs' Profits)} \\
Y_t &= (1-p) A (L_{t-1} + N_{t-1}). && \text{(Aggregate Output)}
\end{aligned}$$

Combining the aggregate balance sheet across all banks $L_t + N_t + M_t = D_t + B_t + E_t$, the government budget constraint, and the household budget constraint yields:

$$C_t^D + C_t^H + L_t + N_t = \left(1 + r_{t-1}^D\right) D_{t-1} - \left(1 + r_{t-1}^B\right) B_{t-1} + E_t - \Xi_t + \tau \Pi_t - \Upsilon_t.$$

Using the expression of the recovery value $\bar{v}_t(l,x,\omega)$, the costs from deposit insurance and bank resolution Υ_t can be rewritten to separate resource cost from revenues from the sale of bank assets:

$$\begin{aligned}
\Upsilon_t &= \int \int_{\bar{\omega}}^{1} \left[\left(1 + r_{t-1}^D\right) d_{t-1}(l,x) + \left(1 + r_{t-1}^B\right) b_{t-1}(l,x) - \bar{v}_t(l,x,\omega)\right] e \, \mathrm{d}F(\omega) \mathrm{d}H_{t-1}(l,x,e) \\
&= \int \int_{\bar{\omega}}^{1} \left[\left(1 + r_{t-1}^D\right) d_{t-1}(l,x) + \left(1 + r_{t-1}^B\right) b_{t-1}(l,x) - \frac{1}{1-\mu} \bar{v}_t(l,x,\omega)\right. \\
&\quad \left. + \frac{\mu}{1-\mu} \bar{v}_t(l,x,\omega)\right] e \, \mathrm{d}F(\omega) \mathrm{d}H_{t-1}(l,x,e) \\
&= \int \int_{\bar{\omega}}^{1} \left[-g_t(l,x,\omega) - \tau \pi_t(l,x,\omega) + \frac{\mu}{1-\mu} \bar{v}_t(l,x,\omega)\right] e \, \mathrm{d}F(\omega) \mathrm{d}H_{t-1}(l,x,e) \\
&= \frac{\mu}{1-\mu} \bar{V}_t - \int \int_{\bar{\omega}}^{1} \left[g_t(l,x,\omega) + \tau \pi_t(l,x,\omega)\right] e \, \mathrm{d}F(\omega) \mathrm{d}H_{t-1}(l,x,e)
\end{aligned}$$

where $\bar{V}_t = \int\int_{\bar{\omega}}^{1} \bar{v}_t(l,x,\omega)e\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,x,e)$.

Combining bank profit taxes with the deposit insurance and bank resolution costs yields:

$$\begin{aligned}
\tau\Pi_t - \Upsilon_t &= \tau\int\int_0^{\bar{\omega}} \pi_t(l,x,\omega)e\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,x,e) \\
&\quad + \int\int_{\bar{\omega}}^{1} \left[g_t(l,x,\omega) + \tau\pi_t(l,x,\omega)\right] e\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,x,e) - \frac{\mu}{1-\mu}\bar{V}_t \\
&= \tau\int\int_0^{1} \pi_t(l,x,\omega)e\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,x,e) \\
&\quad + \int\int_{\bar{\omega}}^{1} g_t(l,x,\omega)e\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,x,e) - \frac{\mu}{1-\mu}\bar{V}_t.
\end{aligned}$$

Combining net equity injections with the equity law of motion yields:

$$\begin{aligned}
E_t - \Xi_t &= (1-\chi)\int\int_0^{\bar{\omega}} g_t(l,x,\omega)e\,dF(\omega)\,dH_t(l,x,e) + \bar{\mathscr{F}}_{t-1}\bar{E}_t \\
&\quad - \left(\bar{\mathscr{F}}_{t-1}\bar{E}_t - \chi\int\int_0^{\bar{\omega}} g_t(l,x,\omega)e\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,x,e)\right) \\
&= \int\int_0^{\bar{\omega}} g_t(l,x,\omega)e\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,x,e).
\end{aligned}$$

Combining the last two expressions yields:

$$\begin{aligned}
E_t - \Xi_t + \tau\Pi_t - \Upsilon_t &= \int\int_0^{\bar{\omega}} g_t(l,x,\omega)e\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,x,e) \\
&\quad + \tau\int\int_0^{1} \pi_t(l,x,\omega)e\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,x,e) \\
&\quad + \int\int_{\bar{\omega}}^{1} g_t(l,x,\omega)e\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,x,e) - \frac{\mu}{1-\mu}\bar{V}_t \\
&= \int\int_0^{1} g_t(l,x,\omega)e\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,x,e) \\
&\quad + \tau\int\int_0^{1} \pi_t(l,x,\omega)e\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,e) - \frac{\mu}{1-\mu}\bar{V}_t \\
&= E_{t-1} + \int\int_0^{1} \pi_t(l,\bar{r},\omega)e\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,\bar{r},e) - \frac{\mu}{1-\mu}\bar{V}_t.
\end{aligned}$$

The double integral over profits in the last expression can be rewritten as follows

$$\begin{aligned}\int\int_0^1 \pi_t(l,x,\omega)e\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,x,e) = (1-p)(\bar{r}_{t-1}^{L*}L_{t-1} + r_{t-1}^{N}N_{t-1}) + r_{t-1}^{M}M_{t-1} \\ - r_{t-1}^{D}D_{t-1} - r_{t-1}^{B}B_{t-1} \\ - \lambda p(L_{t-1}+N_{t-1}) \\ - \int\int_0^1 f\left(\frac{n_{t-1}(l,x)}{l}\right) le\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,x,e) \\ - \bar{\pi}E_{t-1},\end{aligned}$$

where the aggregate rate on legacy loans $\bar{r}_{t-1}^{L*}$ is such that

$$\bar{r}_{t-1}^{L*}L_{t-1} = \int r_{t-1}^{L}(x)l\cdot e\,\mathrm{d}H_{t-1}(l,x,e).$$

Replacing the double integral over profits in the previous expression yields:

$$E_t - \Xi_t + \tau\Pi_t - \Upsilon_t = E_{t-1} - X_{t-1} + \bar{r}_{t-1}^{L*}(1-p)(L_{t-1}+N_{t-1}) + r_{t-1}^{B}B_{t-1} - r_{t-1}^{D}D_{t-1} - RC_t,$$

where $RC_t = \frac{\mu}{1-\mu}\bar{V}_t + \lambda p(L_{t-1}+N_{t-1}) + \int\int_0^1 f\left(\frac{n_{t-1}(l,x)}{l}\right) le\,\mathrm{d}F(\omega)\mathrm{d}H_{t-1}(l,x,e) + \bar{\pi}E_{t-1}$ is the sum of all resource costs in the model.

Substituting this expression into the budget constraint yields

$$C_t^D + C_t^H + L_t + N_t = E_{t-1} - X_{t-1} + D_{t-1} - B_{t-1} + \Pi_t^E + \bar{r}_{t-1}^{L*}(1-p)(L_{t-1}+N_{t-1}) - RC_t.$$

Using the definitions of output and entrepreneur profits, and the balance sheet constraint, we can further simplify the expression

$$C_t^D + C_t^H + L_t + N_t = L_{t-1} + N_{t-1} + Y_t - RC_t,$$

or

$$Y_t = C_t^D + C_t^H + \Delta(L_t + N_t) + RC_t.$$

Thus, output is used for consumption, investment in entrepreneurs' projects, or re-

source cost.

Note that we can express the investments in entrepreneurs' projects as

$$\begin{aligned}\Delta(L_t + N_t) &= L_t + N_t - (L_{t-1} + N_{t-1}) \\ &= (1-p)(1-\chi)(1-\delta)(L_{t-1} + N_{t-1}) + N_t - (L_{t-1} + N_{t-1}) \\ &= N_t - (1-(1-p)(1-\chi)(1-\delta))(L_{t-1} + N_{t-1}),\end{aligned}$$

meaning that it's the amount of new loans made by banks minus the projects that ended regularly, due to project failures, or due to bank exits.

A.5 Bank problem

We start by summarizing the problem of a bank presented in Section 2.1. The problem of a bank is

$$V_t^B(L_{jt}, E_{jt}, x_{jt}^L) = \mathbf{1}_{\{E_{jt} \geq \gamma L_{jt}\}} \left[\max_{\{N_{jt}, M_{jt}, D_{jt}, B_{jt}\}} \beta \mathbb{E}_t[(1-\chi) V_{t+1}^B(L_{jt+1}, E_{jt+1}, x_{jt+1}^L) + \chi E_{jt+1}] \right]$$

$$\begin{aligned}
\text{s.t.} \quad & B_{jt} = L_{jt} + N_{jt} + M_{jt} - D_{jt} - E_{jt}, && \text{(Balance sheet identity)} \\
& D_{jt} \leq \alpha L_{jt}, && \text{(Deposits constraint)} \\
& L_{jt+1} = \left(1 - \omega_{jt+1}\right)(1-\delta)\left(L_{jt} + N_{jt}\right), && \text{(Loan LOM)} \\
& E_{jt+1} = E_{jt} + (1-\tau)\Pi_{jt+1}, && \text{(Equity LOM)} \\
& E_{jt} \geq \gamma\left(L_{jt} + N_{jt}\right), && \text{(Capital requirement)} \\
& M_{jt} \geq \theta(D_{jt} + B_{jt}), && \text{(Reserve requirement)}
\end{aligned}$$

with profits Π_{jt+1} defined as

$$\begin{aligned}\Pi_{jt+1} = (1-\omega_{jt+1})\left(r_{jt}^L L_{jt} + r_t^N N_{jt}\right) + r_t^M M_{jt} - r_t^D D_{jt} - r_t^B B_{jt} \\ - \lambda\omega_{jt+1}(L_{jt} + N_{jt}) - f\left(\frac{N_{jt}}{L_{jt}}\right) L_{jt} - \bar{\pi} E_{jt}.\end{aligned}$$

The state variable x_{jt}^L corresponds to either the loan rate spread on legacy loans s_{jt}^L or the average loan rate on legacy loans r_{jt}^L depending on whether we are in a variable-rate

or fixed-rate economy. We have that

$$r^L_{jt} = \frac{r^L_{jt-1} L_{jt-1} + r^N_{t-1} N_{jt-1}}{L_{jt-1} + N_{jt-1}},$$

for fixed-rate banks, and

$$s^L_{jt} = \frac{s^L_{jt-1} L_{jt-1} + s^N_{t-1} N_{jt-1}}{L_{jt-1} + N_{jt-1}}.$$

for variable-rate banks with $r^L_{jt} = r^M_t + s^L_{jt}$.

The problem above implies a failure threshold $\overline{\omega}_{jt+1}$. If the realization of ω lies above the threshold, a bank fails endogenously

$$\overline{\omega}_{jt+1} = \frac{E_{jt} + (1-\tau)\left[r^L_{jt} L_{jt} + r^N_t N_{jt} + r^M_t M_{jt} - r^D_t D_{jt} - r^B_t B_{jt} - f\left(\frac{N_{jt}}{L_{jt}}\right) L_{jt} - \bar{\pi} E_{jt}\right] - \gamma(1-\delta)(L_{jt} + N_{jt})}{(1-\tau)(r^L_{jt} L_{jt} + r^N_t N_{jt}) + \left[(1-\tau)\lambda - \gamma(1-\delta)\right](L_{jt} + N_{jt})}$$

Reduction in State Variables. Let lower case variables denote ratios of stocks/flows to equity, e.g., $y_{jt} = \frac{Y_{jt}}{E_{jt}}$. The problem of a bank can be written as

$$v^B_t(l_{jt}, x^L_{jt}) = \mathbf{1}_{\{1 \geq \gamma l_{jt}\}} \left[\max_{\{n_{jt}\}} \quad \beta \int_0^{\overline{\omega}_{jt+1}} g_{jt+1} \left[(1-\chi) v^B_{t+1}(l_{jt+1}, x^L_{jt+1}) + \chi \right] \mathrm{d}F(\omega_{jt+1}) \right],$$

$$\begin{aligned}
\text{s.t.} \quad & b_{jt} = l_{jt} + n_{jt} + m_{jt} - d_{jt} - k_{jt}, && \text{(Balance sheet identity)} \\
& d_{jt} = \alpha l_{jt}, && \text{(Deposits constraint)} \\
& l_{jt+1} = (1-\omega_{jt+1}) \frac{(1-\delta)(l_{jt} + n_{jt})}{g_{jt+1}}, && \text{(Loans LOM)} \\
& g_{jt+1} = 1 + (1-\tau)\pi_{jt+1}, && \text{(Equity growth LOM)} \\
& 1 \geq \gamma(l_{jt} + n_{jt}), && \text{(Capital requirement)} \\
& m_{jt} = \theta(d_{jt} + b_{jt}), && \text{(Binding liq. requirement)}
\end{aligned}$$

with

$$\pi_{jt+1} = (1-\omega_{t+1})(r^L_{jt} l_{jt} + r^N_t n_{jt}) + r^M_t m_{jt} - r^B_t b_{jt} - r^D_t d_{jt} - \lambda\omega_{jt+1}(l_{jt}+n_{jt}) - f\left(\frac{n_{jt}}{l_{jt}}\right) l_{jt} - \overline{\pi},$$

$$\overline{\omega}_{jt+1} = \frac{1+(1-\tau)\left[r^L_{jt} l_{jt} + r^N_t n_{jt} + r^M_t m_{jt} - r^D_t d_{jt} - r^B_t b_{jt} - f\left(\frac{n_{jt}}{l_{jt}}\right) l_{jt} - \bar{\pi}\right] - \gamma(1-\delta)(l_{jt}+n_{jt})}{(1-\tau)(r^L_{jt} l_{jt} + r^N_t n_{jt}) + \left[(1-\tau)\lambda - \gamma(1-\delta)\right](l_{jt}+n_{jt})}.$$

A bank's decisions, therefore, only depends on its leverage l_t and on the average loan rate spread on legacy loans s^L_{jt} for variable-rate banks and the average loan rate on legacy loans r^L_t for fixed-rate banks, respectively.

A.6 Proof of Proposition 1

We start with the problem of the fixed-rate bank. We guess and verify later that, in the absence of idiosyncratic risk, the regulatory constraint is never binding (and thus the bank never fails). In that case, the problem of a bank j is:

$$\max_{\{N_{j,t}, M_{j,t}, D_{j,t}, B_{j,t}\}_{t=0}^{\infty}} \quad \sum_{t=0}^{\infty} \beta^{t+1}(1-\chi)^t \chi E_{j,t+1}, \tag{45}$$

subject to

$$L_{j,t+1} = (1-p)(1-\delta)(L_{j,t}+N_{j,t}), \tag{46}$$

$$E_{j,t+1} = E_{j,t} + (1-\tau)\Pi_{j,t+1}, \tag{47}$$

$$r^L_{j,t+1} = \frac{r^L_{j,t} L_{j,t} + r^N_t N_{j,t}}{L_{j,t}+N_{j,t}}, \tag{48}$$

$$\begin{aligned}\Pi_{j,t+1} = (1-p)\left(r^L_{j,t} L_{j,t} + r^N_t N_{j,t}\right) + r^M_t M_{j,t} - r^D_t D_{j,t} - r^B_t B_{j,t} \\ - \lambda p(L_{j,t}+N_{j,t}) - f\left(N_{j,t}/L_{j,t}\right) L_{j,t},\end{aligned} \tag{49}$$

$$L_{j,t} + N_{j,t} + M_{j,t} = D_{j,t} + B_{j,t} + E_{j,t}, \tag{50}$$

$$D_{j,t} \le \alpha L_{j,t}, \tag{51}$$

$$M_{j,t} \ge \theta(B_{j,t} + D_{j,t}). \tag{52}$$

Assuming the deposit constraint (51) and the liquidity requirement (52) are always binding, the choice variables $D_{j,t}$, $M_{j,t}$, and $B_{j,t}$ can be expressed as functions of the single choice variable for new loans, $N_{j,t}$, and the state variables $L_{j,t}$ and $E_{j,t}$.

The bank's problem is reduced to:

$$\max_{\{N_{j,t}\}_{t=0}^{\infty}} \quad \sum_{t=0}^{\infty} \beta^{t+1}(1-\chi)^t \chi E_{j,t+1}, \tag{53}$$

subject to the laws of motion:

$$L_{j,t+1} = (1-p)(1-\delta)(L_{j,t} + N_{j,t}), \tag{54}$$

$$E_{j,t+1} = E_{j,t} + (1-\tau)\Pi_{j,t+1}, \tag{55}$$

$$r^L_{j,t+1} = \frac{r^L_{j,t} L_{j,t} + r^N_t N_{j,t}}{L_{j,t} + N_{j,t}}, \tag{56}$$

where bank profits, $\Pi_{j,t+1}$, are given by:

$$\begin{aligned}\Pi_{j,t+1} = (1-p)\left(r^L_{j,t} L_{j,t} + r^N_t N_{j,t}\right) + r^M_t \left[\frac{\theta}{1-\theta}(L_{j,t} + N_{j,t} - E_{j,t})\right] \\ - r^D_t \alpha L_{j,t} - r^B_t \left[\frac{(1-\alpha(1-\theta))L_{j,t} + N_{j,t} - E_{j,t}}{1-\theta}\right] \\ - \lambda p(L_{j,t} + N_{j,t}) - f\left(N_{j,t}/L_{j,t}\right) L_{j,t}.\end{aligned} \tag{57}$$

We can now iterate forward the laws of motion of the state variables:

Stock of loans $L_{j,t+k}$. To find the expression for $L_{j,t+k}$ as a function of $L_{j,t}$ and the sequence of new loans $\{N_{j,t+m}\}_{m=0}^{k-1}$, we can solve the law of motion for loans by forward iteration. The law of motion for $L_{j,t+k}$ is given by:

$$L_{j,t+1} = (1-p)(1-\delta)(L_{j,t} + N_{j,t}).$$

Let's define a constant $\varrho = (1-p)(1-\delta)$. The equation becomes $L_{j,t+1} = \varrho(L_{j,t} + N_{j,t})$. Then, for any time $t+k$:

$$L_{j,t+k} = \varrho N_{j,t+k-1} + \varrho^2 N_{j,t+k-2} + \cdots + \varrho^{k-1} N_{j,t+1} + \varrho^k N_{j,t} + \varrho^k L_{j,t},$$

which can be written more compactly as:

$$L_{j,t+k} = \varrho^k L_{j,t} + \sum_{m=0}^{k-1} \varrho^{k-m} N_{j,t+m}.$$

Average loan rate $r^L_{j,t+k}$. To find the expression for $r^L_{j,t+k}$, we start with its law of motion:

$$r^L_{j,t+1} = \frac{r^L_{j,t} L_{j,t} + r^N_t N_{j,t}}{L_{j,t} + N_{j,t}}.$$

We can define the bank j's total interest income from legacy loans in period t as $I_{j,t} = r^L_{j,t} L_{j,t}$. From the law of motion for $r^L_{j,t+1}$:

$$r^L_{j,t+1}(L_{j,t} + N_{j,t}) = r^L_{j,t} L_{j,t} + r^N_t N_{j,t}.$$

We know from the law of motion for loans that $L_{j,t+1} = \varrho(L_{j,t} + N_{j,t})$. Then $L_{j,t} + N_{j,t} = L_{j,t+1}/\varrho$. Substituting this into the rearranged equation gives:

$$r^L_{j,t+1} L_{j,t+1} = \varrho(r^L_{j,t} L_{j,t} + r^N_t N_{j,t}).$$

This gives us a law of motion for the total interest income, $I_{j,t+1} = \varrho(I_{j,t} + r^N_t N_{j,t})$. We can solve this by forward iteration, similar to how we solved for $L_{j,t}$:

$$I_{j,t+k} = r^L_{j,t+k} L_{j,t+k} = \varrho^k r^L_{j,t} L_{j,t} + \sum_{m=0}^{k-1} \varrho^{k-m} r^N_{t+m} N_{j,t+m}.$$

To find the expression for $r^L_{j,t+k}$, we simply divide the total interest income, $r^L_{j,t+k} L_{j,t+k}$, by the total loan stock, $L_{j,t+k}$. Using the expression for $L_{j,t+k}$ from the previous step we get:

$$r^L_{j,t+k} = \frac{\varrho^k r^L_{j,t} L_{j,t} + \sum_{m=0}^{k-1} \varrho^{k-m} r^N_{t+m} N_{j,t+m}}{\varrho^k L_{j,t} + \sum_{m=0}^{k-1} \varrho^{k-m} N_{j,t+m}}.$$

Equity $E_{j,t+k+1}$. The expression for bank equity $E_{j,t+k+1}$ in period $t+k+1$ is a function of the initial states $(E_{j,t}, L_{j,t}, r^L_{j,t})$, the history of the bank's choices for new loans $(\{N_{j,t+m}\}_{m=0}^{k-1})$, the path of exogenous interest rates $(\{r^M_{t+m}, r^B_{t+m}, r^D_{t+m}\}_{m=0}^{k-1})$, and the path of endogenous new loan rates $(\{r^N_{t+m}\}_{m=0}^{k-1})$.

The law of motion for bank equity is given by:

$$E_{j,t+1} = E_{j,t} + (1-\tau)\Pi_{j,t+1}.$$

The profit function $\Pi_{j,t+1}$ can be written compactly as:

$$\Pi_{j,t+1} = (1-p)I_{j,t} + \Phi_t L_{j,t} + \Psi_t N_{j,t} - f\left(\frac{N_{j,t}}{L_{j,t}}\right) L_{j,t} + \mu_t E_{j,t}, \tag{58}$$

where

$$\begin{aligned}
\Phi_t &\equiv \alpha(r_t^B - r_t^D) - \lambda p - \mu_t, \\
\Psi_t &\equiv (1-p) r_t^N - \lambda p - \mu_t, \\
\mu_t &\equiv \frac{r_t^B - \theta r_t^M}{1-\theta}.
\end{aligned}$$

The law of motion of equity can now be rewritten as:

$$E_{j,t+1} = \Upsilon_t E_{j,t} + (1-\tau)\left[(1-p)I_{j,t} + \Phi_t L_{j,t} + \Psi_t N_{j,t} - f\left(\frac{N_{j,t}}{L_{j,t}}\right) L_{j,t}\right], \tag{59}$$

where $\Upsilon_t \equiv \left[1 + (1-\tau)\mu_t\right]$. By iterating this relationship forward from t to $t+k+1$, we obtain the final expression:

$$\begin{aligned}
E_{j,t+k+1} = &\left(\prod_{m=0}^{k} \Upsilon_{t+m}\right) E_{j,t} \\
&+ (1-\tau) \sum_{m=0}^{k} \left(\prod_{q=m+1}^{k} \Upsilon_{t+q}\right) \Big[(1-p)I_{j,t+m} + \Phi_{t+m} L_{j,t+m} \\
&\qquad + \Psi_{t+m} N_{j,t+m} - f\left(\frac{N_{j,t+m}}{L_{j,t+m}}\right) L_{j,t+m}\Big],
\end{aligned} \tag{60}$$

which holds with the convention that an empty product is equal to one ($\prod_{q=k+1}^{k} \Upsilon_{t+q} = 1$).

First-order condition for $N_{j,t}$. To obtain the first-order condition for the bank's optimal choice of new loans, we differentiate the objective function with respect to $N_{j,t}$. The objective function is:

$$V_{j,t} = \sum_{k=0}^{\infty} \beta^{k+1} (1-\chi)^k \chi E_{j,t+k+1}.$$

Using the expression for $E_{j,t+k+1}$ derived previously, we can isolate the terms depending on $N_{j,t}$. The choice of $N_{j,t}$ affects the equity path through two channels: the direct impact on profits in period t (where $m = 0$), and the persistent impact on the stocks of loans and interest income in all future periods $t + m$ (where $m \geq 1$). We define the effective discount factor for a unit of equity realized in period $t + m + 1$ as $\Lambda_{t,m}$:

$$\Lambda_{t,m} \equiv (1-\tau) \sum_{k=m}^{\infty} \beta^{k+1} (1-\chi)^k \chi \left(\prod_{q=m+1}^{k} \Upsilon_{t+q} \right). \tag{61}$$

This term represents the present value of the expected dividends generated by a marginal unit of after-tax profit in period $t + m$. Differentiating the objective function requires summing the impact of $N_{j,t}$ on the profit flow at each horizon m, weighted by $\Lambda_{t,m}$.

1. Immediate impact ($m = 0$): The choice of $N_{j,t}$ directly affects profits at t through the net return on new loans Ψ_t and the adjustment costs.

$$\frac{\partial \Pi_{j,t+1}}{\partial N_{j,t}} = \Psi_t - f'\left(\frac{N_{j,t}}{L_{j,t}}\right).$$

2. Future impact ($m \geq 1$): The choice of $N_{j,t}$ increases the stock of loans and interest income in future periods. From the laws of motion derived earlier, the marginal effect of $N_{j,t}$ on future states is:

$$\frac{\partial L_{j,t+m}}{\partial N_{j,t}} = \varrho^m,$$
$$\frac{\partial I_{j,t+m}}{\partial N_{j,t}} = r_t^N \varrho^m.$$

The impact on the profit flow at time $t + m$ is:

$$\begin{aligned}\frac{\partial \Pi_{j,t+m+1}}{\partial N_{j,t}} &= (1-p)\frac{\partial I_{j,t+m}}{\partial N_{j,t}} + \Phi_{t+m}\frac{\partial L_{j,t+m}}{\partial N_{j,t}} - \left[f'\left(\frac{N_{j,t+m}}{L_{j,t+m}}\right)\frac{N_{j,t+m}}{L_{j,t+m}} - f\left(\frac{N_{j,t+m}}{L_{j,t+m}}\right)\right]\frac{\partial L_{j,t+m}}{\partial N_{j,t}} \\ &= \varrho^m \left[(1-p) r_t^N + \Phi_{t+m} + f'\left(\frac{N_{j,t+m}}{L_{j,t+m}}\right)\frac{N_{j,t+m}}{L_{j,t+m}} - f\left(\frac{N_{j,t+m}}{L_{j,t+m}}\right)\right].\end{aligned}$$

Combining these terms, the expression for the derivative of the objective function with

respect to $N_{j,t}$ is:

$$\frac{\partial V_{j,t}}{\partial N_{j,t}} = \Lambda_{t,0}\left[\Psi_t - f'\left(\frac{N_{j,t}}{L_{j,t}}\right)\right] + \sum_{m=1}^{\infty}\Lambda_{t,m}\varrho^m\left[(1-p)r_t^N + \Phi_{t+m} + f'\left(\frac{N_{j,t+m}}{L_{j,t+m}}\right)\frac{N_{j,t+m}}{L_{j,t+m}} - f\left(\frac{N_{j,t+m}}{L_{j,t+m}}\right)\right]. \quad (62)$$

Variable-rate bank. We can proceed in a similar way to obtain the expression that characterizes the optimal choice of $N_{j,t}$ for the variable-rate bank:

$$\frac{\partial V_{j,t}}{\partial N_{j,t}} = \Lambda_{t,0}\left[\tilde{\Psi}_t - f'\left(\frac{N_{j,t}}{L_{j,t}}\right)\right] + \sum_{m=1}^{\infty}\Lambda_{t,m}\varrho^m\left[(1-p)s_t^N + \tilde{\Phi}_{t+m} + f'\left(\frac{N_{j,t+m}}{L_{j,t+m}}\right)\frac{N_{j,t+m}}{L_{j,t+m}} - f\left(\frac{N_{j,t+m}}{L_{j,t+m}}\right)\right], \quad (63)$$

where the auxiliary variables are defined as:

$$\tilde{\Phi}_t \equiv \alpha(r_t^B - r_t^D) - \lambda p - \mu_t + (1-p)r_t^M,$$
$$\tilde{\Psi}_t \equiv (1-p)(s_t^N + r_t^M) - \lambda p - \mu_t.$$

Optimal new loans $N_{j,t}$ (Fixed-Rate Bank). Setting the derivative $\frac{\partial V_{j,t}}{\partial N_{j,t}} = 0$, substituting the functional form for origination costs, and solving for new loans $N_{j,t}$ we obtain:

$$N_{j,t} = \frac{L_{j,t}}{2\eta}\left[\Psi_t + \sum_{m=1}^{\infty}\frac{\Lambda_{t,m}}{\Lambda_{t,0}}\varrho^m\left((1-p)r_t^N + \Phi_{t+m} + \eta\left(\frac{N_{j,t+m}}{L_{j,t+m}}\right)^2\right)\right]. \quad (64)$$

Optimal new loans $N_{j,t}$ (Variable-Rate Bank). Setting the derivative $\frac{\partial V_{j,t}}{\partial N_{j,t}} = 0$, substituting the functional form for origination costs, and solving for new loans $N_{j,t}$ we obtain:

$$N_{j,t} = \frac{L_{j,t}}{2\eta}\left[\tilde{\Psi}_t + \sum_{m=1}^{\infty}\frac{\Lambda_{t,m}}{\Lambda_{t,0}}\varrho^m\left((1-p)s_t^N + \tilde{\Phi}_{t+m} + \eta\left(\frac{N_{j,t+m}}{L_{j,t+m}}\right)^2\right)\right]. \quad (65)$$

Relationship between fixed rate r_t^N and spread s_t^N. Equating the optimal choices of new loans for the fixed- and variable-rate banks implies that the fixed rate r_t^N must

equal the fixed spread s_t^N plus a weighted average of the current and future policy rates $\{r_{t+m}^M\}_{m=0}^{\infty}$. Defining the weighting factor for horizon m as $w_{t,m} \equiv \frac{\Lambda_{t,m}}{\Lambda_{t,0}}\varrho^m$ (with $w_{t,0} = 1$), the relationship is:

$$r_t^N = s_t^N + \frac{\sum_{m=0}^{\infty} w_{t,m} r_{t+m}^M}{\sum_{m=0}^{\infty} w_{t,m}}. \tag{66}$$

This condition states that for the bank to be indifferent between the two pricing structures, the fixed rate must "price in" the expected future path of the market rates that the bank gives up by locking in a fixed rate at origination. The weights depend on the loan survival rate (ϱ) and the bank's effective discount factor (Λ).

Relationship between r_t^N and s_t^N from loan demand. We seek the relationship between the fixed rate r_t^N and the variable spread s_t^N that ensures the quantity of new loans demanded, N_t, is identical in both economies. We equate the demand functions (12) and (13) and let Ω_m be the effective discount factor for the entrepreneur's payoffs m periods after the loan origination (specifically, the interest payment at $t+m+1$):

$$\Omega_m \equiv \beta^{m+1}(1-p)^{m+1}(1-\delta)^m(1-\tilde{\chi})^m.$$

The demand equality condition is:

$$\sum_{m=0}^{\infty} \Omega_m (A - r_t^N) = \sum_{m=0}^{\infty} \Omega_m \left[A - (r_{t+m}^M + s_t^N)\right].$$

Since A, r_t^N, and s_t^N are known at time t and do not depend on m, we can separate the terms:

$$A \sum_{m=0}^{\infty} \Omega_m - r_t^N \sum_{m=0}^{\infty} \Omega_m = A \sum_{m=0}^{\infty} \Omega_m - \sum_{m=0}^{\infty} \Omega_m r_{t+m}^M - s_t^N \sum_{m=0}^{\infty} \Omega_m.$$

The terms involving the parameter A cancel out. We can rearrange the remaining terms to isolate r_t^N:

$$r_t^N = s_t^N + \frac{\sum_{m=0}^{\infty} \Omega_m r_{t+m}^M}{\sum_{m=0}^{\infty} \Omega_m}. \tag{67}$$

This result mirrors the supply-side condition: for the entrepreneur to be indifferent between fixed and variable rate loans, the fixed rate must equal the fixed spread plus a weighted average of the expected future path of market rates.

Final result. For the irrelevance result to hold, we would need:

$$\frac{\sum_{m=0}^{\infty} w_{t,m} r_{t+m}^{M}}{\sum_{m=0}^{\infty} w_{t,m}} = \frac{\sum_{m=0}^{\infty} \Omega_m r_{t+m}^{M}}{\sum_{m=0}^{\infty} \Omega_m}. \tag{68}$$

Recall that:

$$\begin{aligned}
w_{t,m} &\equiv \frac{\Lambda_{t,m}}{\Lambda_{t,0}} \varrho^m, \\
\Lambda_{t,m} &\equiv (1-\tau) \sum_{k=m}^{\infty} \beta^{k+1} (1-\chi)^k \chi \left(\prod_{q=m+1}^{k} \Upsilon_{t+q} \right), \\
\Upsilon_t &\equiv \left[1 + (1-\tau)\mu_t \right], \\
\mu_t &\equiv \frac{r_t^B - \theta r_t^M}{1-\theta}, \\
\varrho &\equiv (1-p)(1-\delta),
\end{aligned}$$

and

$$\Omega_m \equiv \beta^{m+1} (1-p)^{m+1} (1-\delta)^m (1-\tilde{\chi})^m.$$

A.7 Law of motion of a bank's equity

Banks can either fail endogenously ($\omega > \bar{\omega}$), or exit exogenously ($\iota = 1$) with probability χ. Since the realization of ω and ι are independent, there are four possible cases, which we handle as follows:

$$\begin{aligned}
&\omega > \bar{\omega} \text{ and } \iota = 1 \quad \rightarrow \text{Exogenous Exit (Bank resolution mechanism)}, \\
&\omega \leq \bar{\omega} \text{ and } \iota = 1 \quad \rightarrow \text{Exogenous Exit (Regular)}, \\
&\omega > \bar{\omega} \text{ and } \iota = 0 \quad \rightarrow \text{Endogenous Failure}, \\
&\omega \leq \bar{\omega} \text{ and } \iota = 0 \quad \rightarrow \text{Continues Operating}.
\end{aligned}$$

A bank's equity at $t+1$ is a function of states $(l_{jt}, x^L_{jt}, e_{jt})$ at time t and shocks (ω, ι) at $t+1$.[26] We can write the law of motion of equity as

$$\begin{aligned} e_{jt+1}(l_{jt}, x^L_{jt}, e_{jt}, \omega, \iota) = {} & \mathbf{1}_{\{\omega \le \bar{\omega}, \iota = 0\}} g_{t+1}(l_{jt}, x^L_{jt}, \omega) e_{jt} \\ & + \mathbf{1}_{\{\omega > \bar{\omega}, \iota = 0\}} \bar{E}_{t+1} \\ & + \mathbf{1}_{\{\omega \le \bar{\omega}, \iota = 1\}} \bar{E}_{t+1} \\ & + \mathbf{1}_{\{\omega > \bar{\omega}, \iota = 1\}} \bar{E}_{t+1} . \end{aligned}$$

Due to the independence of ω and ι, we can rewrite this as

$$\begin{aligned} e_{jt+1}(l_{jt}, x^L_{jt}, e_t, \omega, \iota) = {} & \mathbf{1}_{\{\omega \le \bar{\omega}\}} \mathbf{1}_{\{\iota = 0\}} g_{t+1}(l_{jt}, x^L_{jt}, \omega) e_{jt} \\ & + \left[\mathbf{1}_{\{\omega > \bar{\omega}\}} \mathbf{1}_{\{\iota = 0\}} + \mathbf{1}_{\{\omega \le \bar{\omega}\}} \mathbf{1}_{\{\iota = 1\}} + \mathbf{1}_{\{\omega > \bar{\omega}\}} \mathbf{1}_{\{\iota = 1\}} \right] \bar{E}_{t+1} , \end{aligned}$$

where

$$g_{t+1}(l_{jt}, x^L_{jt}, \omega) = 1 + (1-\tau)\pi_{jt+1}(l_{jt}, x^L_{jt}, \omega),$$

denotes the gross equity growth rate in the case a bank operates successfully ($\omega \le \bar{\omega}$).

Integrating over the Bernoulli distribution for ι yields

$$\begin{aligned} \int_0^1 e_{jt+1}(l_{jt}, x^L_{jt}, e_t, \omega, \iota) dX(\iota) = {} & (1-\chi) \mathbf{1}_{\{\omega \le \bar{\omega}\}} g_{t+1}(l_{jt}, x^L_{jt}, \omega) e_{jt} \\ & + \left[\mathbf{1}_{\{\omega > \bar{\omega}\}} + \chi \mathbf{1}_{\{\omega \le \bar{\omega}\}} \right] \bar{E}_{t+1} . \end{aligned}$$

Integrating over ω yields

$$\begin{aligned} \int_0^1 \int_0^1 e_{jt+1}(l_{jt}, x^L_{jt}, e_{jt}, \omega, \iota) dX(\iota) dF(\omega) = {} & (1-\chi) \int_0^{\bar{\omega}} g_{t+1}(l_{jt}, x^L_{jt}, \omega) e_{jt} dF(\omega) \\ & + \left(1 - F(\bar{\omega}) + \chi F(\bar{\omega})\right) \bar{E}_{t+1} \\ = {} & (1-\chi) \bar{g}_{t+1}(l_{jt}, x^L_{jt}) e_{jt} + \left[1 - (1-\chi) F(\bar{\omega})\right] \bar{E}_{t+1}, \end{aligned}$$

where $\bar{g}_{t+1}(l_{jt}, x^L_{jt}) = \int_0^{\bar{\omega}} g_{t+1}(l_{jt}, x^L_{jt}, \omega) dF(\omega)$. Note that $\bar{\omega}$ itself is a function of leverage l_t and the average loan rate/spread x^L_{jt}, which for notational simplicity we have omitted.

Then, integrating over the joint distribution of leverage, the average loan rate/spread,

[26] Since the mass of banks is constant, we are treating newly entering banks after endogenous failures and exogenous exits as direct successors of the failing bank.

and equity $H_t(l_{jt}, x^L_{jt}, e_{jt})$ yields

$$E_{t+1} = (1-\chi)G_t E_t + \bar{\mathscr{F}}_t \bar{E}_{t+1},$$

where the aggregate gross equity growth rate G_t and the mass of banks exiting or failing $\bar{\mathscr{F}}_t$ are given by

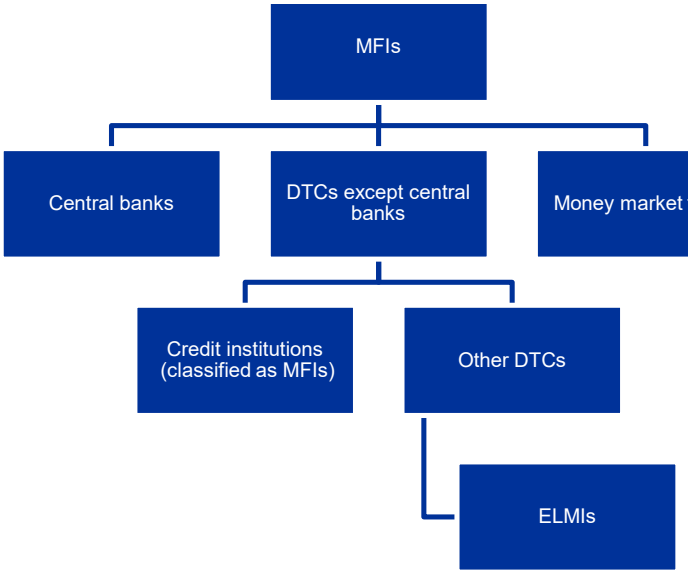
$$G_t = \frac{1}{E_t}\int \bar{g}_{t+1}(l_{jt}, x^L_{jt}) e_{jt} dH_t(l_{jt}, x^L_{jt}, e_{jt}), \text{ and}$$
$$\bar{\mathscr{F}}_t = \int \left[1-(1-\chi)F(\bar{\omega})\right] dH_t(l_{jt}, x^L_{jt}, e_{jt}).$$

B. Empirical Appendix

B.1 Consolidated banks balance sheet: Data sources

This section explains how we map the Consolidated Balance Sheet of the Euro Area MFIs (excluding the Eurosystem) to the banks' balance sheet in the model.

Figure 10: Components of the MFIs sector



Note: DTC stands for deposit-taking corporation. ELMI stands for electronic money institution.

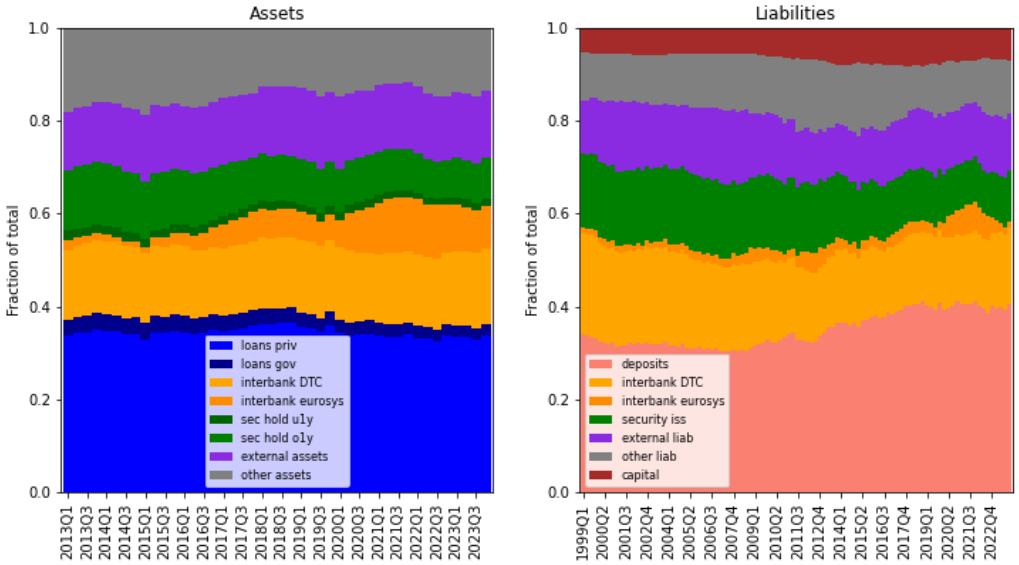
The main source is the Statistical Data Warehouse (SDW) of the ECB. We use monthly or quarterly data subject to availability and transform the series to quarterly frequency. The period of analysis starts corresponds to 01/1999 - 01/2024. Datasets:

- Consolidated balance sheet of the MFIs (excluding the Eurosystem). https://data.ecb.europa.eu/publications/money-credit-and-banking/3031821
- MFI holdings of securities breakdown by maturity and types: Debt securities, equity, and non-MMF investment fund shares. https://data.ecb.europa.eu/publications/money-credit-and-banking/3031889
- Sectoral breakdown of MFI loans vis-à-vis the private sector. https://data.ecb.europa.eu/publications/money-credit-and-banking/3031822

B.2 Euro Area MFIs Balance Sheet

This section summarizes the salient features of the aggregated balance sheet of monetary financial institutions (MFIs) operating in the euro area, excluding the Eurosystem.[27] These MFIs include deposit-taking institutions (banks), and money market funds (MMFs).

Figure 11: MFIs consolidated balance sheet in the euro area, 2013-2023



Source: ECB SDW. Aggregated Balance Sheet of Euro Area Monetary Financial Institutions (MFIs) excluding the Eurosystem. MFIs are comprised of deposit-taking corporations, money market funds, and central banks.

An inspection of the asset side of Monetary and Financial Institutions (MFIs) in the euro area from 1999 to 2023, in Figure 11, shows that their asset composition has been remarkably stable. On average, the lending portfolio to households, firms, and the government accounts for 62% of assets. Interbank loans—which include repurchase agreements (repos), securities lending, and similar operations with other MFIs and national central banks—account for about 15% of assets. Security holdings, both short and long-term, account for the remaining 23%.[28] On the liabilities side, most liabilities

[27]The Eurosystem includes the European Central Bank (ECB) and the national central banks of the countries of the euro area.

[28]We assign external assets and other assets proportionally to the loans and short-term security holdings

(60%) are deposits and interbank deposits (17%). The remaining liabilities are securities (16 %) and capital (7%).[29] See the breakdown below in the Table 4.

Table 4: MFIs balance sheet composition (2013–2023)

Assets		**Liabilities**	
Loans	0.62	Deposits	0.63
Interbank loans	0.15	Interbank deposits	0.15
Short-term security holdings	0.12	Security issuance	0.14
Long-term security holdings	0.11	Capital	0.08

Source: ECB Statistical Data Warehouse. Aggregated Balance Sheet of Euro Area MFIs, excluding the Eurosystem. Time series averages across periods. Loans: include loans to the private sector, loans to government, a fraction (85%) of external assets (i.e. operations with non-euro area residents) and other assets. Interbank loans: includes interbank loans with other DTCs. Short-term security holdings: include security holdings with a maturity of less than a 1 year plus interbank operations with NCBs (repos and security lending). Long-term security holdings: include security holdings with a maturity greater than 1 year. Deposits: include retail deposits of different maturities, external liabilities with non-euro area residents, and other liabilities. Interbank deposits refer to interbank deposits with other DTCs. Security Issuance includes the issuance of short and long-term securities plus interbank operations with NCBs.

B.3 CET 1 Capital Ratios and Buffers

Table 5: Average capital ratios for euro area banks

	All banks	Large	Supervised
CET 1 capital ratio	15.62	13.23	14.45
CET 1 management buffer ratio	7.97	6.16	5.12

Note: All numbers are in percentage points. The first two columns correspond to the cross-sectional means of the centered distribution grouped from 2013 to 2020. *All banks* refers to all euro area banks in our sample, approximately 70+ per quarter. *Large banks* refer to banks with total assets larger than 100 billion euros. *Supervised banks* refer to 64 banks supervised by the European Banking Authority (EBA) in 2021.Q4. *Sources:* Regulatory requirements (GSII, OSI, SRB) are obtained from the European Systemic Risk board (ESRB). Data for the Pillar 2 requirements of CET1 capital from ECB supervisory data. Bank-level data for CET 1 ratios and Total Risk Weighted Assets from *S&P* Global.

categories. External assets are holdings of cash in currencies other than the euro, holdings of securities issued by non-residents of the euro area, and loans to non-residents of the euro area (including banks). For statistical purposes, these items are included indistinguishably in MFIs' external assets without identifying them separately.

[29]Notice that this aggregate measure of capital and reserves does not coincide with the regulatory capital that we present in Appendix B.3, which is the Core Equity Tier 1 (CET1) capital expressed as a percentage of risk-weighted assets.

Table 5 reports the cross-sectional average CET1 capital ratios and capital buffers for different types of euro area banks.

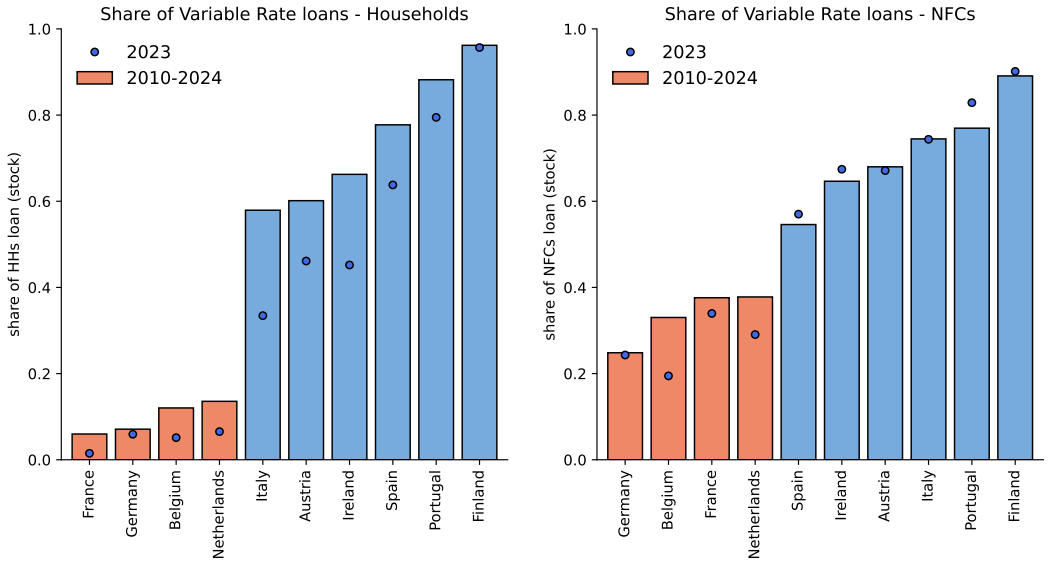
The first two columns present the cross-sectional averages of the grouped distribution for a sample of euro area banks from 2013 to 2020. We construct an unbalanced bank-level panel using balance-sheet data from S&P Global, a proprietary source. Our quarterly dataset includes information on common equity tier 1 (CET1) capital levels, risk-weighted assets, and total assets. For each bank in the sample, we calculate the capital buffer as the difference between its CET1 ratio and the applicable Combined Buffer Requirement (CBR) in each quarter. The CBR is defined as the sum of the Capital Conservation Buffer (CCoB), the Countercyclical Capital Buffer (CCyB), and the maximum of the following institution-specific components: the Systemic Risk Buffer (SRB), the Global Systemically Important Institution (G-SII) buffer, and the Other Systemically Important Institution (O-SII) buffer.

The third column presents CET1 ratios and capital buffer estimates for a sample of Supervised Banks as of 2021:Q4 reported by the European Banking Authority (EBA). These estimates incorporate Pillar 2 requirements for CET1 capital in addition to the combined buffer requirements. As expected, the average CET1 capital buffer is slightly lower once Pillar 2 requirements are included. Nonetheless, the overall distribution retains a similar shape and statistical properties.

B.4 Loan Rate Pricing

Figure 12 shows the share of variable rate lending broken down by credit to households and non-financial corporations. The bars display the average for 2010Q1-2024Q1, which, for most countries, is close to the average for 2023 (blue circles), suggesting persistence in loan rate pricing practices.

Figure 12: Composition of lending stocks by interest rate fixation period.



Data sources: ECB Statistical Data Warehouse. The left panel presents the share of the stock of aggregate lending to households (including mortgage loans, consumer loans, and other loans) issued at variable rates. The right panel presents the share of stock of aggregate lending to non-financial corporations issued at variable rates. The bars display the average for 2010Q1-2024Q1. Orange bars corresponds to our classification of fixed-rate countries and blue bars to variable-rates. Blue circles depict the average for the year 2023.

B.5 Estimating Local Projections

We estimate Local Projections a là Jordà (2005) and Jordà et al. (2015). We use as a monetary policy shock the first differences in the deposit facility rate (DFR) instrumented with a measure of monetary surprises.

We built a balanced panel for the twenty euro are countries. In our baseline estimations, presented in the paper, we restrict to the ten largest countries (Austria, Belgium, Germany, Finland, France, Italy, Ireland, Netherlands, Portugal, and Spain) because this allows to construct a balanced panel without data gaps. Variables include lending,

deposit rates, and net interest margin (NIM) rates, lending volumes, capital/equity ratios, and macroeconomic indicators (inflation rates, GDP growth, employment, among others) from 2000 to 2023. All variables in the panel are consolidated at the country level, the data frequency is quarterly.

Countries are categorized as variable-raters (VR) if their share of variable-rate lending is above 50% or fixed-raters (FR) otherwise. VR countries are Spain, Portugal, Italy, Finland, Ireland, and Austria. FR countries are Germany, France, Belgium, and the Netherlands. In Appendix B.6 we present robustness checks for the extended sample with all 20 euro area countries, and for a restricted sample where we exclude a set of southern european countries. All IRFs remain qualitatively unchanged.

Interest rates. We estimate the following local projection specification:

$$r_{c,t+h} = \alpha_{c,h} + \beta_{1h}\varepsilon_t^{MP} + \beta_{2h}\left[\varepsilon_t^{MP} \times I_c^{FR}\right] + \Gamma_h X_{c,t} + e_{c,t+h} \tag{69}$$

where $r_{c,t+h}$ denotes the variable of interest (lending rates, deposit rates, NIM rates) at time t, and horizon h ranging from 0 to 16 quarters. The variable ε_t^{MP} denotes the monetary policy shock at time t, which we instrument– in a first stage– with the *(median) monetary policy component* from Jarociński and Karadi (2020). I_c^{FR} is a dummy variable that takes the value of one when a country belongs to the FR category.

$X_{c,t}$ represents the set of controls. As it is common in the literature, we include the first lag of the dependent variable and the first lag of the deposit facility rate as controls. We also use the contemporaneous and the first lag of inflation and the quarterly growth rate of the industrial production index. As well as the first lag of the yield on a BBB corporate bond index for the euroarea, and the first lag of the yield on the one-year German government debt bond since these variables have been found relevant for the euro area (Jarociński and Karadi (2020)).

Quantities. In a similar fashion, our econometric specification for the volume of lending:

$$\log Y_{c,t+h} = \alpha_{c,h} + \beta_{1h}\epsilon_t^{MP} + \beta_{2h}\left[\epsilon_t^{MP} \times I_c^{FR}\right] + \Gamma_h X_{c,t} + e_{c,t+h}. \tag{70}$$

For these specifications, we use the monetary surprises (i.e., the *(median) monetary policy component*, ϵ_t^{MP}, from Jarociński and Karadi (2020)) without instrumenting the DFR since for log-volumes the monetary surprise series yields smoother IRFs. The set

of controls is the same used for interest rates but expressing variables in logarithms: first lag of the dependent variable $\log Y_{c,t-1}$. The contemporaneous and the first lag of HICP and the log of the industrial production index. We also include the first lag of the yield on a BBB corporate bond index for the euroarea, and the first lag of the yield on the one-year German government debt bond.

Figure 4 (in the main text) shows the estimated IRFs for interest rates, and lending volumes. The sample period is 2003 to 2019.[30] In each Figure, the left Panel depicts the sequence of estimated dynamic coefficients $\{\beta_{1h}\}_{h=0}^{h=16}$ of the monetary policy shock as a solid blue line together with 95% confidence bands. This represents the average effect across VR countries. The right Panel depicts the sequence of estimated dynamic coefficients $\{\beta_{1h} + \beta_{2h}\}_{h=0}^{h=16}$ of the monetary policy shock as a solid blue line together with 95% confidence bands. This represents the average effect across FR countries.

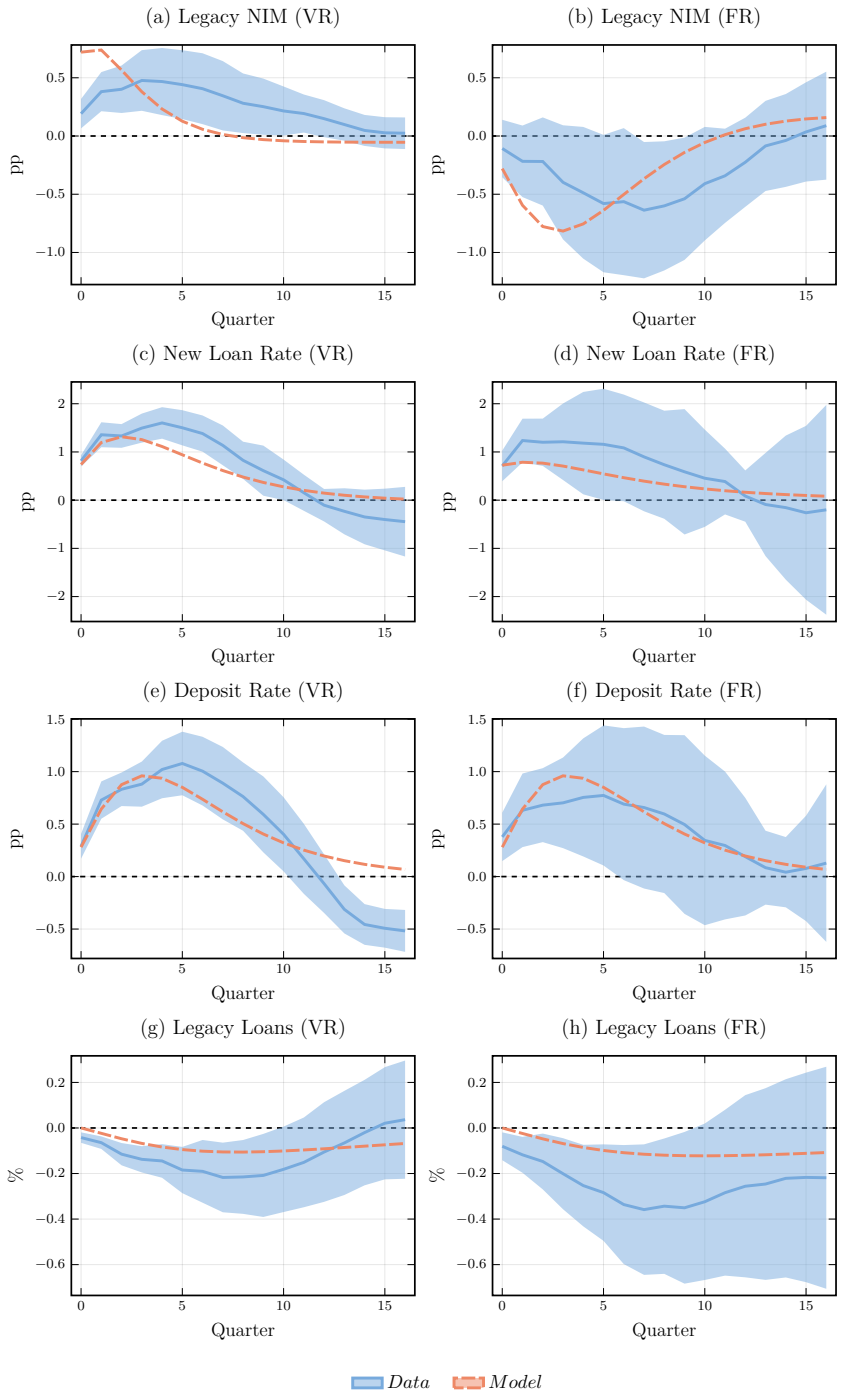
B.6 Robustness for Local Projections

Panel of 20 euro area countries. Figure 13 presents robustness estimations for an extended sample including all 20 euro area countries, where Germany, France, Belgium, Netherlands, and Slovakia are classified as operating under a fixed rate regime (FR). Portugal, Spain, Finland, Ireland, Austria, Italy, Estonia, Croatia, Cyprus, Greece, Latvia, Lithuania, Malta, Luxembourg, and Slovenia are classified as operating under a variable rate regime (VR).

Panel of 20 euro area countries excluding periphery countries. Figure 14 shows that our estimated empirical responses are not driven by a core-periphery classification. We present robustness estimations for the extended sample of 20 euro area countries but excluding a set of periphery countries: Spain, Portugal, Italy and Ireland. This exclusion reduces the set of countries categorized as VR to Finland, Austria, Estonia, Croatia, Cyprus, Greece, Latvia, Lithuania, Malta, Luxembourg, and Slovenia. The set of FR countries is Germany, France, Belgium, Netherlands, and Slovakia.

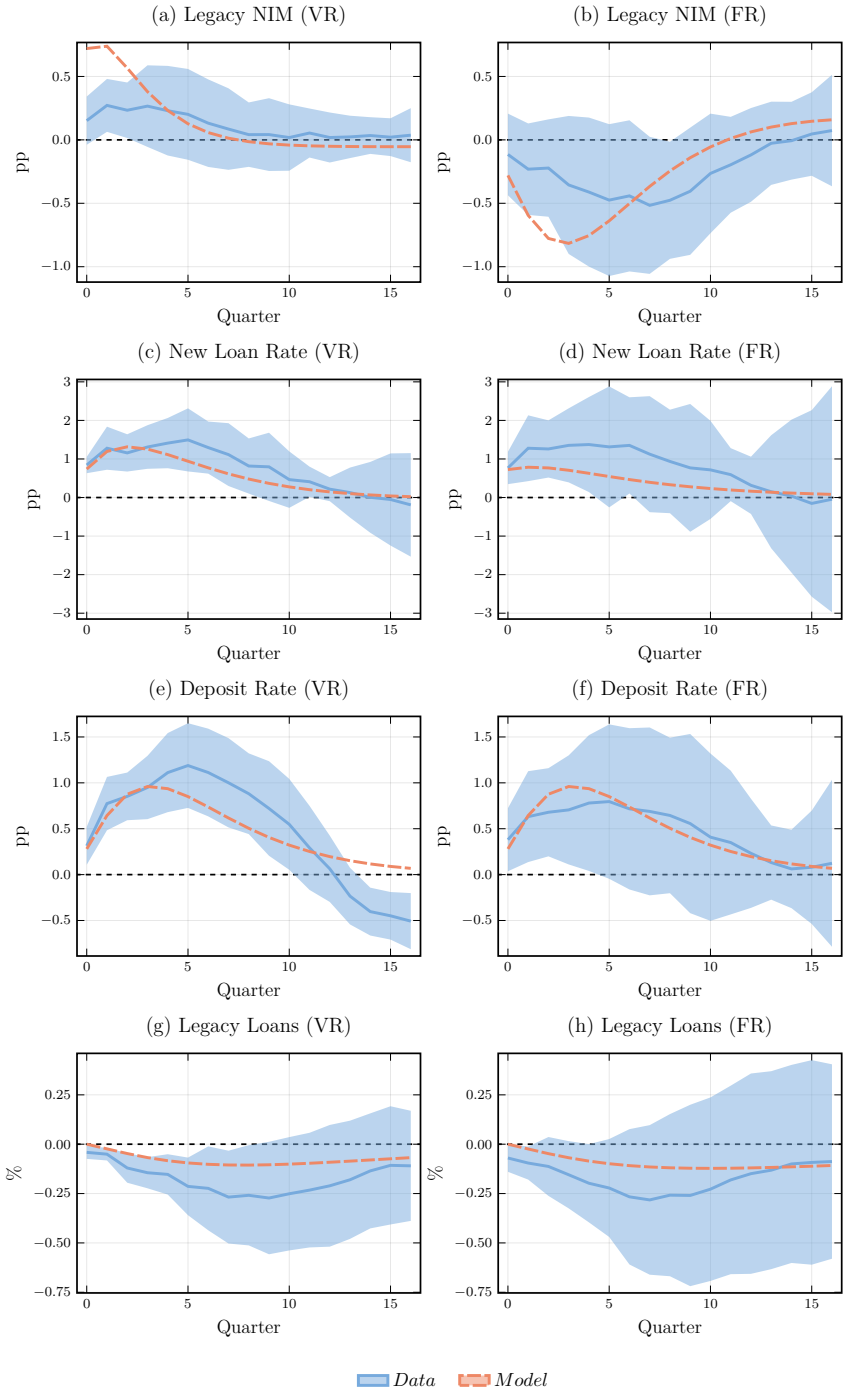
[30] The estimates for the sample period 2003 to 2023 are qualitatively similar.

Figure 13: Local Projections. Panel of 20 euro area countries.



Note: Solid blue lines show the empirical impulse responses to a monetary policy shock and dashed red lines compute the model counterparts. Light blue bands show the 95% confidence intervals. Panels on the left report the response across VR countries, while right Panels report the response across FR countries in the data and in the model. See Appendix B.5 for estimation details.

Figure 14: Local Projections. Panel of 20 euro area countries excluding periphery countries.

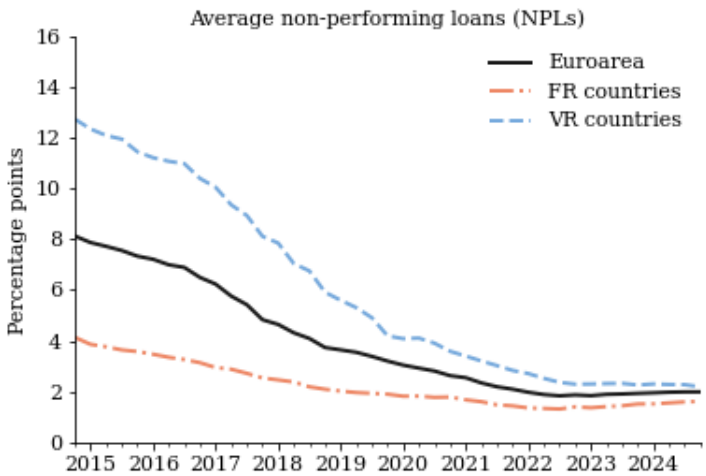


Note: Solid blue lines show the empirical impulse responses to a monetary policy shock and dashed red lines compute the model counterparts. Light blue bands show the 95% confidence intervals. Panels on the left report the response across VR countries, while right Panels report the response across FR countries in the data and in the model. See Appendix B.5 for estimation details.

B.7 Credit risk in the euro area

This section examines credit risk dynamics across euro area countries using time-series data on non-performing loans (NPLs) from the ECB's Consolidated Banking Data (CBD2) dataset. The dataset covers the period from 2014 Q1 to 2024 Q1. As before, we focus on the ten largest euro area economies, grouped by their share of variable-rate lending. FR countries: Belgium, France, Germany, and the Netherlands, and VR countries: Austria, Finland, Ireland, Italy, Portugal, and Spain.

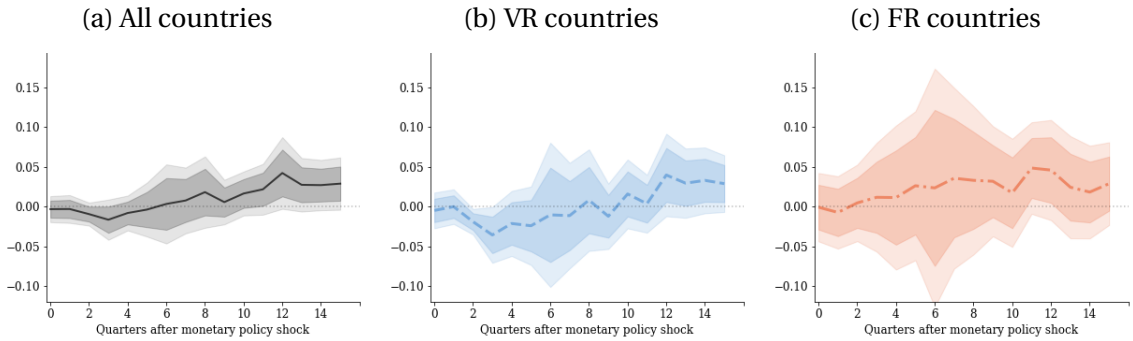
Figure 15: Average non-performing loans (NPLs).



Note: NPLs are defined as the volume of non-performing loans and advances divided by the total volume of loans and advances. The data is from the European Central Bank's country *Consolidated Banking Data* (CBD2) dataset. The country sample covers the ten largest euro area economies, grouped by their share of variable-rate lending. FR countries: Belgium, France, Germany, and the Netherlands, and VR countries: Austria, Finland, Ireland, Italy, Portugal, and Spain.

Figure 15 shows the average ratio of NPLs— defined as the volume of non-performing loans and advances divided by the total volume of loans and advances —across FR, VR, and all euro area countries. There are clear differences in levels across country groups: VR countries feature historically higher NPLs than their FR counterparts. However, over the last decade, VR countries have seen their NPL ratios decline, converging to levels similar to those seen for FRs. Using granular credit registry data, Core et al. (2025) documents similar patterns in non-financial corporation's default rates across fixed- and variable-rate economies in the Euro Area.Bandoni et al. (2025) document a similar

Figure 16: Local Projections on NPLs



Note: NPLs are defined as the volume of non-performing loans and advances divided by the total volume of loans and advances. The data is from the European Central Bank's country *Consolidated Banking Data* (CBD2) dataset. The country sample covers the ten largest euro area economies, grouped by their share of variable-rate lending. FR countries: Belgium, France, Germany, and the Netherlands, and VR countries: Austria, Finland, Ireland, Italy, Portugal, and Spain.

declining pattern for mortgage default rates computed from a sample of securitized mortgages in Spain, Portugal, Ireland, and Italy.[31]

Figure 16 shows the responses of NPLs to a monetary surprise estimated by Local Projections. We estimate the same specification used for interest rates, see Appendix B.5. Panel (a) shows the average response across all countries. Panels (b) and (c) show the responses for VR and FR countries, respectively. Responses do not differ significantly across country group types.[32]

[31] Core et al. (2025) use Anacredit loan level data in December 2021, they find that the 1-year probability of default (PD) on fixed-rate loans averaged 6.26% with a 19.64% standard deviation, and 9.52% with a standard deviation of 25.05% for variable-rate loans. Bandoni et al. (2025) use loan-level data on securitised mortgages from the European DataWarehouse (EDW). For the period 2014 to 2019, they estimate an average mortgage default rate of 0.9% across countries.

[32] Our estimates capture the responses of the credit risk stock, since NPLs are a slow-moving measure of credit risk (stock). Understanding the responses of flow measures, such as defaults at 60 or 90 days, would require using granular credit registry data, as no country-consolidated series are available.

C. Solution Algorithm

Preliminaries. For the solution algorithm, we define a new choice variable

$$k_t^{gap} = 1 - \gamma(l_t + n_t)\,,$$

which captures how slack the capital constraint is. Note that given this definition, the capital constraint simplifies to $k_t^{gap} \geq 0$. Using the choice variable k_t^{gap} and the state l_t, we can then compute n_t as

$$n_t = \frac{1 - \gamma l_t - k_t^{gap}}{\gamma}$$

Given the expression for n_t, all other model variables can be computed using the expressions presented in the main text and the appendix.

The solution algorithm then aims to find a policy function for k_t^{gap} that maximizes the value function such that $k_t^{gap}, n_t, d_t, b_t, m_t \geq 0$. Note that the constraint on d_t is always satisfied since $l_t \geq 0$ and $d_t = \alpha l_t$. If, additionally, $b_t \geq 0$ then the constraint on m_t is also satisfied since $m_t = \theta(b_t + d_t)$. Therefore, we only need to ensure

$$\begin{aligned} n_t &= \frac{1 - \gamma l_t - k_t^{gap}}{\gamma} \geq 0\,, \\ b_t &\frac{l_t + n_t + m_t + (\theta - 1) d_t - 1}{} \geq 0\,, \end{aligned}$$

which implies two upper bounds on k_t^{gap}

$$\begin{aligned} 1 - \gamma l_t &\geq k_t^{gap}\,, \\ 1 - \gamma(1 + (1-\theta)\alpha l_t) &\geq k_t^{gap}\,. \end{aligned}$$

The constraints define a maximum feasible value for k_t^{gap}

$$k_t^{gap,max} = \min\{1 - \gamma l_t,\, 1 - \gamma(1 + (1-\theta)\alpha l_t)\}\,,$$

In the implementation of Algorithm 1, it is ensured that these constraints are not violated.

Steady State. Solving for the model's steady state comprises two main steps: First, solving for the individual bank policy functions using value function iteration. Second,

computing the steady-state bank distribution over equity E_t, leverage l_t, and the average loan rate/spread x_t^L using the method of Young (2010). These steps must then be executed iteratively to find the equilibrium loan rate r^L which clears the loan market.

We discretize the state space for $l_t \in [0, 1/\gamma]$, $x_t^L \in [x^L - \sigma, x^L + \sigma]$, where σ is the size of the MIT shock, and $\log(E_t) \in [\log(0.13), \log(3000)$ using equally spaced grids.[33] Algorithm 1 describes the value function iteration algorithm used to solve the problem of an individual bank which, due to size-independence, only depends on (l_t, x_t^L). Algorithm 2 describes the algorithm to compute the bank distribution. Finally, Algorithm 3 describes the complete algorithm used to solve for the steady state.

Algorithm 1 (Individual Problem)**.**

1. *Make a guess for the capital gap policy function $k_0^{gap}(l, x^L)$ and the value function $V_0(l, x^L)$.*

2. *Taking the value function for next period $V_i(l, x^L)$ as given, use an optimization routine to find the value of $k_{i+1}^{gap}(l, x^L)$ that maximizes todays value $V_{i+1}(l, x^L)$ for each grid point (l, x^L). Note that we use cubic interpolation to interpolate the value function if (l_{t+1}, x_{t+1}^L) are off-grid.*

3. *Optional "Howard Improvement": Keeping the capital gap policy function $k_{i+1}^{gap}(l, x^L)$, update the value function by iterating on it N times.*

Iterate on steps 2 & 3 until the maximum absolute difference between $V_{i+1}(l, x^L)$ and $V_i(l, x^L)$ is less than a given degree of precision.

Algorithm 2 (Bank Distribution)**.**

1. *Make a guess for the bank distribution $H(l, x^L, \log(E))$ in the form of a matrix $\mathscr{H}_0$ where each element corresponds to the mass associated with a particular grid point $(l, x^L, \log(E))$.*

2. *Given the individual policy function and the distribution $\mathscr{H}_i$, determine the closest grid points to which banks move in the next period and redistribute mass using the method of Young (2010) yielding $\mathscr{H}_{i+1}$.*

[33] Note that, technically, there is no need for the x^L grid in the steady state, since it stays constant for all banks. However, computing the steady-state policies for $x_t^L \neq x^L$ is required when computing the transition after an MIT shock, such that bank behavior is well-defined when interest rates are back at their steady-state value, even if the average loan rate/spread at individual banks x_t^L has not yet converged back to the steady state.

Iterate on steps 2 until the maximum absolute difference between $\mathscr{H}_{i+1}$ and $\mathscr{H}_i$ is less than a given degree of precision.

Algorithm 3 (Steady State)**.**

1. *Make an initial guess for the loan rate r^N.*
2. *Solve the individual bank problem as described in Algorithm 1.*
3. *Solve for the bank distribution as described in Algorithm 2.*
4. *Check whether r^N clears the loan market. If the loan market does not clear, update the guess for r^N and go to step 2.*

Transition. We use an algorithm similar to the one described in Boppart et al. (2018) to solve for the transitional dynamics after an MIT shock. The approach is similar in spirit to the steady-state algorithm presented above. However, in this case, we are not trying to find a single value for the loan rate r^L to clear the loan market but a path $\{r_t^N\}_{t=0}^T$ to clear the loan market in each period.

Algorithm 4 (Transition)**.**

1. *Choose a time T at which the economy is assumed to have reached the steady state.*
2. *Guess a path for the loan rate $\{r_t^N\}_{t=0}^T$.*
3. *Solve the value and policy functions backward from $t = T-1, \ldots, 1$ assuming that time T value and policy functions correspond to the ones in the steady state.*[34]
4. *Update the paths for the distribution $\{\mathscr{H}_t(l, x^L, \log(E))\}_{t=0}^T$ by iterating forwards from $t = 1, \ldots, T$ using the updated path of policy functions from the previous step.*
5. *Given the path for the distribution, the policy functions, and the loan demand schedule, compute the implied path for the loan rate.*
6. *Compute the maximum difference between the implied paths for $\{r_t^N\}_{t=0}^T$ and its guess. Stop the algorithm if the maximum difference is less than a given degree of precision.*

[34] This part of the algorithm proceeds analogously to solving for the steady state.

7. *Update the guess* $\{r_t^N\}_{t=0}^T$ *by taking a weighted average of the old guess and the implied paths. Go to step 3.*